

MAXIMO ENTERPRISE ASSET MANAGEMENT AT OTORITA IBU KOTA NEGARA (OIKN)

Training Material – Inventory Management

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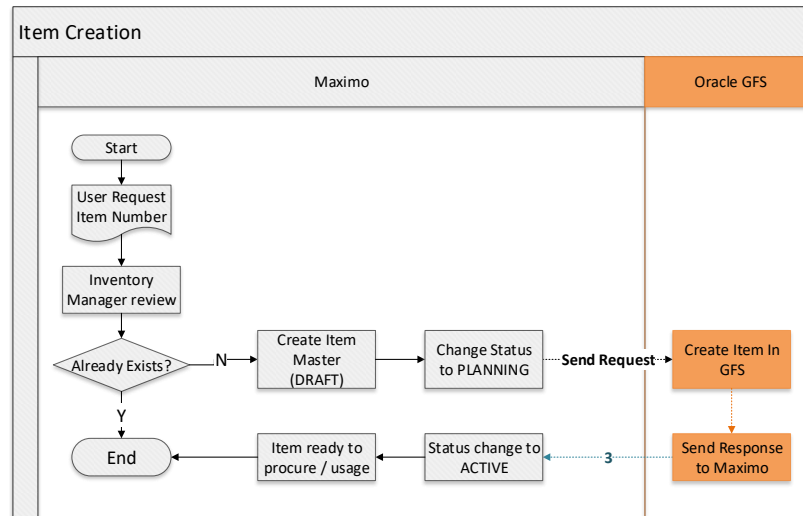
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1 Item, Service, and Tool

1.1 Overview

Item, Service, and Tool are catalogue of items that usually purchased on behalf of Maintenance work. Each of them have different functions, which are:

1. Item
Item is type of item that consumed in Maintenance work. Spare parts and It Inventory belongs to this type of item. This type of item usually is stored in Storeroom.
2. Service
Service is service type item, purchased when a Maintenance work needs service from outside parties. Frequent purchased service should be catalogued in EMS.
3. Tool
Tool is type of item that supports Maintenance work. The notable differences between **Items** and **Tools** are the items consumed, while tool is not. Tool can be stored in Storeroom too. Tool usage in maintenance work won't be charged.



Item, Service, and Tool master data related to Engineering spare-parts and IT inventory data will be created in EMS. The procedure shown above described how item in EMS created:

1. End user need to create Item Master Data Request manually.
2. Engineering Manager will review whether the requested item master already exists or not in EMS.
3. The result of review process done by Engineering Manager are follows:
 - If the item master requested already exists, Engineering Manager will inform to user the requested item number.
 - If the item master requested is not existed yet, Engineering Manager will create item master data and input the details information about this item.

4. Created item needs to be integrated to Oracle GFS, because the procurement process will involve Oracle GFS system. After integration occurred and succeed, the item can be received in inventory or used in maintenance work.

There are some status values that can be applied on an item master data, to track the status and accessibility on the item. Those values are as follows:

- **DRAFT** : Initial status of the item when the item is created.
- **PLANNING** : User change the status of item into **PLANNING** to perform integration process to Oracle GFS. Changing details of the **ACTIVE** item will automatically change the status into **PLANNING** too, to initiate update integration to Oracle GFS automatically.
- **ACTIVE** : Status of item that can be received or used in maintenance work. When the integration process is complete, system will change the status into **ACTIVE** automatically.
- **PENDOBS** : Item with this status cannot be received or issued from inventory. User change the status in **PENDOBS** when user plans to deactivate this item. User still can reactivate this item by changing status back to **ACTIVE**, or deactivate it by changing the status to **OBSOLETE**.
- **OBSOLETE** : Indicates that the item is deactivated, and cannot be activated again.

Stocked items and tools will be saved in storeroom. Each of storerooms will have different GL account to record every transactions against the storeroom. There are two storerooms created in EMS which are:

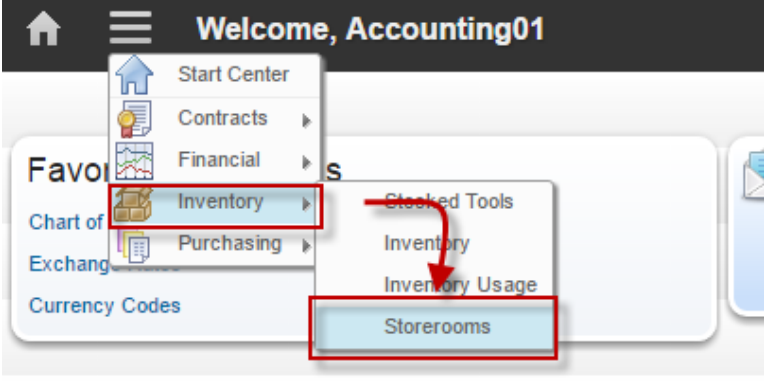
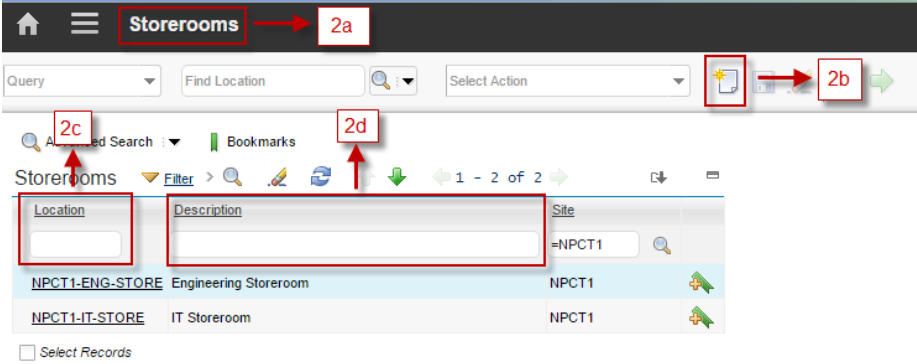
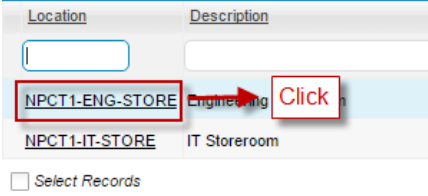
1. **COMP1-ENG-STORE**: This storeroom contains all spare-parts and tools for engineering department.
2. **COMP1-IT-STORE**: This storeroom contains all IT materials that has been purchased for IT Department.

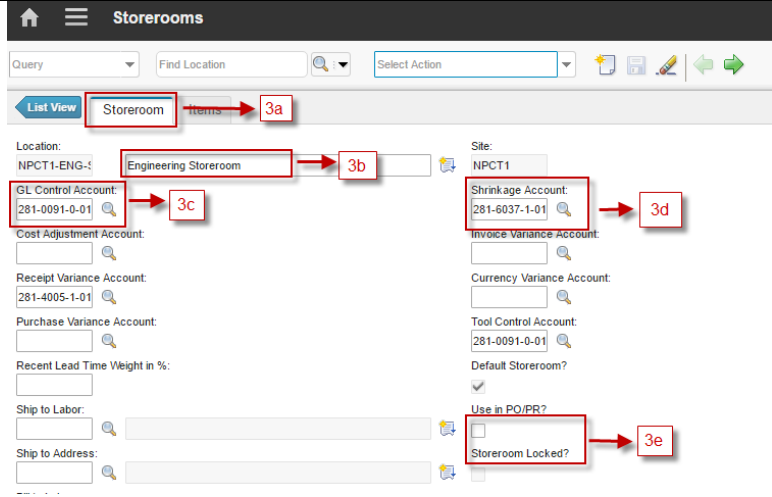
Each item can be imbued with additional specification based on classification where those item is belongs to. By defining the specification and its value, this can help user to determine which item is the most suitable for a maintenance work. Example of specification that can be added to an item such as *Maximum Voltage Reading* (for Multimeter), or *Power* (for Generator Set). Specification master data can be accessed via **Classification** application that is discussed in **Work Management (part 1)** training material.

1.2 Storerooms Application

In **Storeroom** application, user can add or maintain information about storeroom locations, as well as view the items stocked within a storeroom. User also use this application to associate the general ledger accounts with each storeroom, or lock/unlock a storeroom when stock opname performed toward those storeroom.

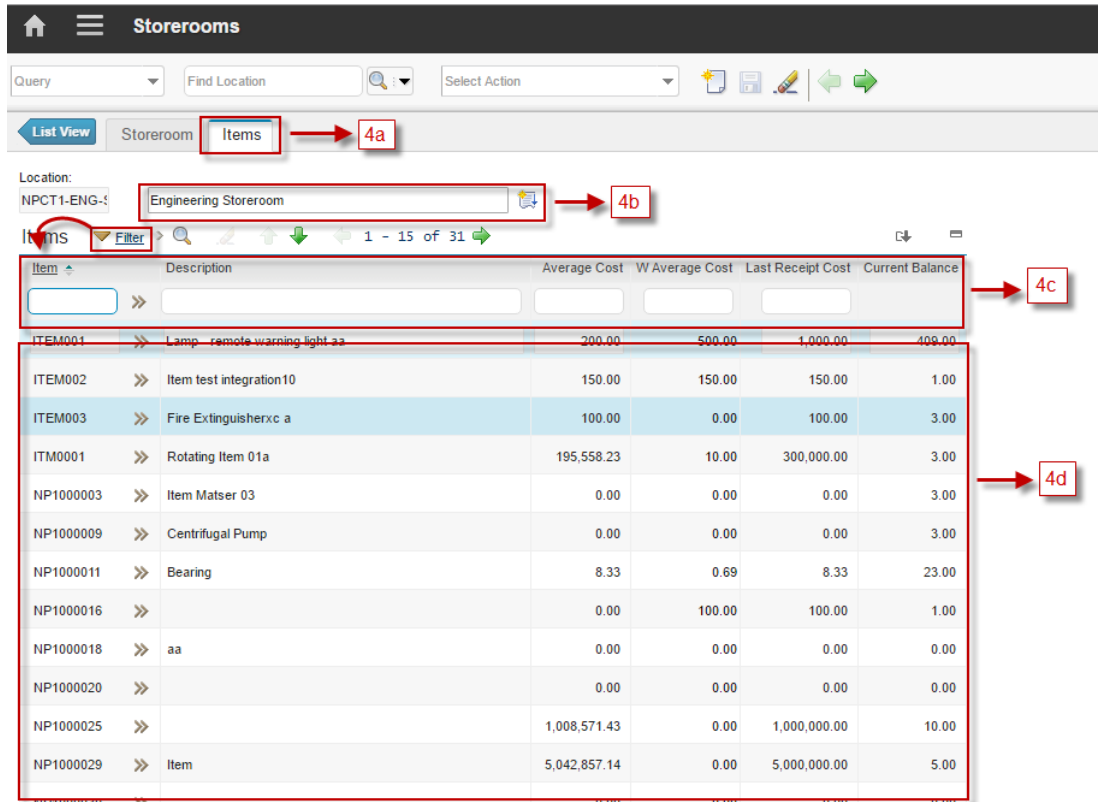
Following are details about Storeroom Applications accessibility:

1	 From the Go To menu, choose Inventory > Storerooms
2	 2.a Header Bar, give information that user is in the Storerooms application. 2.b New Storerooms button, to add new storerooms. 2.c Location field, to filter storerooms by location's code. 2.d Description field, to filter storerooms by location's description.
3	 Choose one storeroom to show the details about the storeroom. Will show display like below picture.

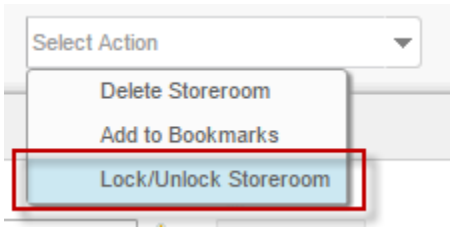


3.a In **Storeroom** tab, user can see detail information about storeroom
 3.b Information about description of the storeroom
 3.c Information about GL Control Account of the storeroom
 3.d Information about Shrinkage Account of the storeroom
 3.e Information about lock status of the storeroom.

4 Choose Tab **Items** in the beside Storeroom tab (4a).

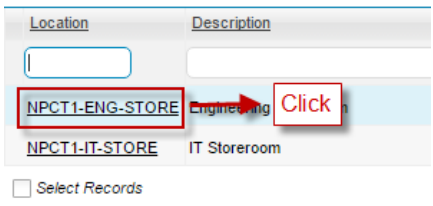
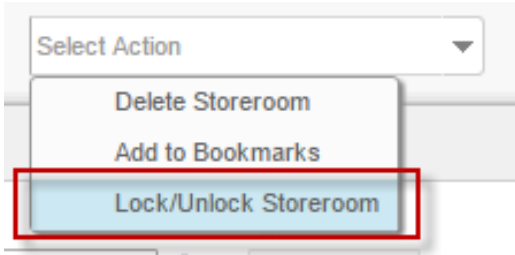


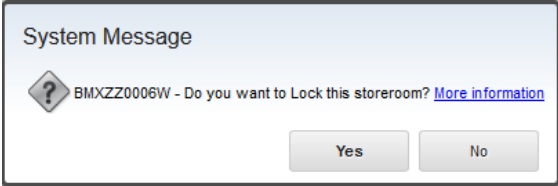
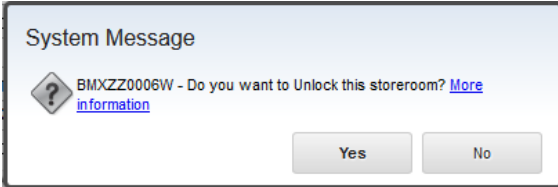
Item	Description	Average Cost	W Average Cost	Last Receipt Cost	Current Balance
ITEM001	Lamp-remote warning light aa	200.00	500.00	1,000.00	408.00
ITEM002	Item test integration10	150.00	150.00	150.00	1.00
ITEM003	Fire Extinguisherxc a	100.00	0.00	100.00	3.00
ITM0001	Rotating Item 01a	195,558.23	10.00	300,000.00	3.00
NP1000003	Item Matser 03	0.00	0.00	0.00	3.00
NP1000009	Centrifugal Pump	0.00	0.00	0.00	3.00
NP1000011	Bearing	8.33	0.69	8.33	23.00
NP1000016		0.00	100.00	100.00	1.00
NP1000018	aa	0.00	0.00	0.00	0.00
NP1000020		0.00	0.00	0.00	0.00
NP1000025		1,008,571.43	0.00	1,000,000.00	10.00
NP1000029	Item	5,042,857.14	0.00	5,000,000.00	5.00

	4.a In this tab, user can see all items stored in the storeroom 4.b Information about description of the storeroom 4.c Filter item to look for specific items in storeroom. Filtering can be done based on Item Code, Description, Average Cost, or Last Receipt Cost. 4.d All items in storeroom as filtering result.
5	User can perform some action related to the storeroom from Select Action menu (on top of the form).  Lock/Unlock Storeroom : Action to change lock status of the storeroom.

1.2.1 Lock/Unlock Storeroom

Lock/Unlock storeroom is used to prevent or grant an inventory transaction toward the storeroom. This feature is useful in Stock opname, to prevent stock movement while the Stock opname is performed. This feature can be performed by *Accounting*. Below are steps to lock/unlock a storeroom:

1	From Storeroom application, choose the storeroom that wants to be locked. 
2	On Storeroom tab, notify that the Storeroom Locked is unchecked. Storeroom Locked? ☐
3	From Select Action menu, click Lock/Unlock Storeroom. 

	A system message will appears. Click Yes.  Notify that Storeroom Locked is checked now. Storeroom Locked? ☒
4	To unlock the storeroom, from Select Action menu, click Lock/Unlock Storeroom again. A system message will appears. Click Yes.  Notify that Storeroom Locked is unchecked now.

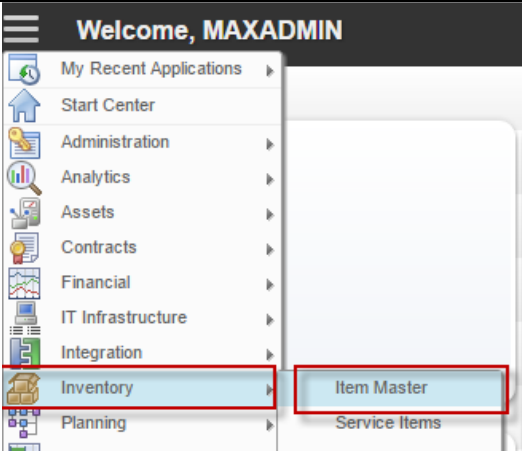
1.3 Item Master Application

Item Master application used to add or maintain item master data with the type of **Item**. There are some important key terms that should be understood regarding **Item Master** application in EMS:

- **Item Set**
EMS establishes items at enterprise level so that multiple organizations can use them. Cost, vendor information, and so forth differ among the organizations, but the overall item definition list (item master) can be shared between those organizations. Therefore, items are created in an item set. When user is creating an organization, specify the item set that this organization will use. Only one item set can be specified for an organization. However, multiple organizations can use the same item set. This item set is applied to **Service** and **Tool** type item too.
- **Rotating**
Rotating means that each of the item will be treated as asset. Therefore, each of them will have an asset record on their own in **Asset** Application.
Here is a consideration to help deciding whether an item master needs to be defined as rotating or not. When user need to track each of the item's movement and maintenance, user need to tag an asset number toward each of them, so user set the item as rotating. Else, if user don't care about the item's movement or maintenance, user set the item as non-rotating.

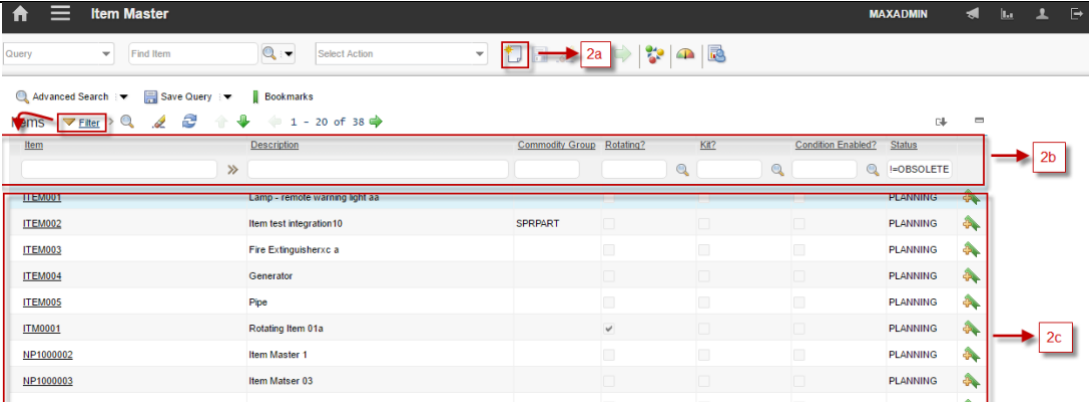
Followings are details about **Item Master** applications accessibility:

1



From **Go To** menu, choose **Inventory – Item Master**

2



Header Bar will give information that user is in the Item Master application.

2.a **New Item** button, to add new item

2.b **Filter** items by **Item Number**, **Description**, **Commodity Group**, **Rotating**, **Kit**, **Condition Enable**, and **Status**.

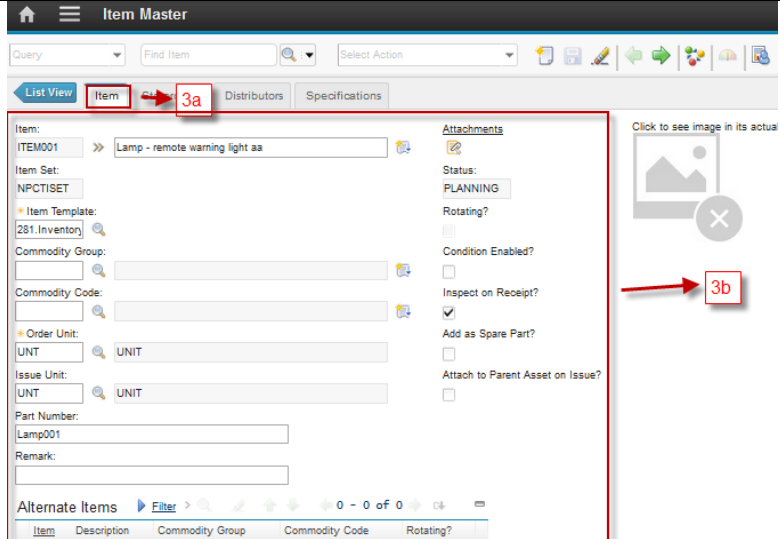
2.c List of all items based of filtering result.

3

Choose one item, to show the details of the item

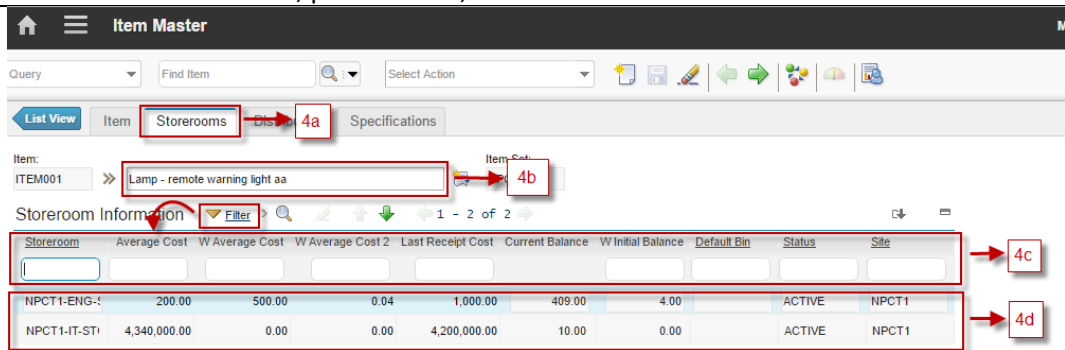


Will show display like below picture.



3.a **Item** tab, to view and update details of item

3.b Detail information of the item, such as item set, item template, commodity group and codes, Order and issue unit, part number, etc.



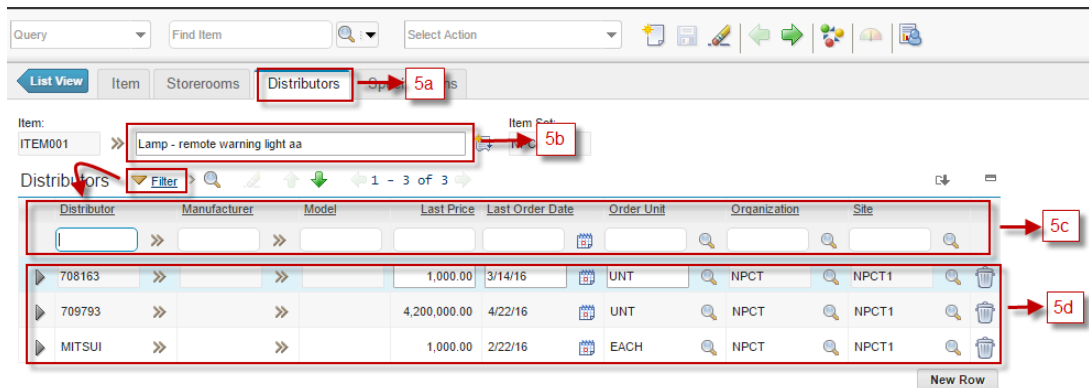
Storeroom	Average Cost	W Average Cost	W Average Cost 2	Last Receipt Cost	Current Balance	W Initial Balance	Default Bin	Status	Site
NPCT1-ENG-1	200.00	500.00	0.04	1,000.00	409.00	4.00		ACTIVE	NPCT1
NPCT1-IT-ST	4,340,000.00	0.00	0.00	4,200,000.00	10.00	0.00		ACTIVE	NPCT1

4.a **Storerooms** tab, to view all storerooms related to this item.

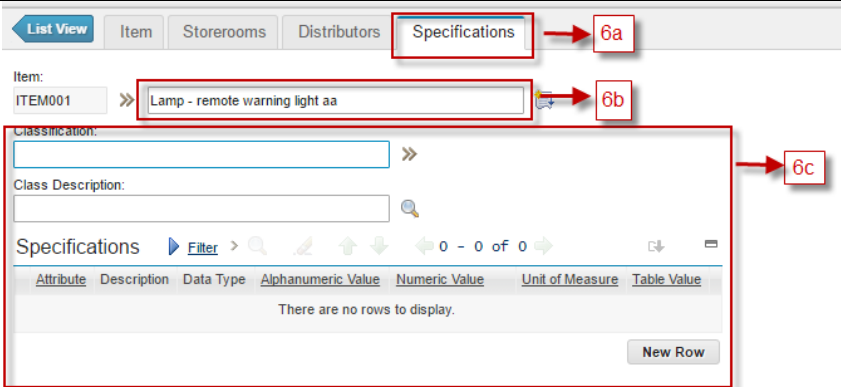
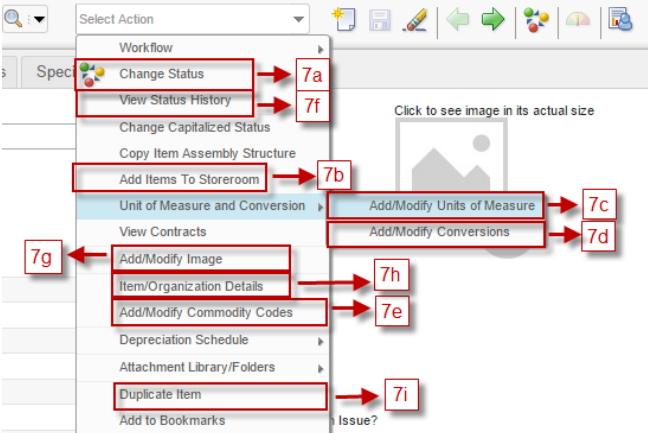
4.b Information about the item's description.

4.c Filter storerooms by **Storeroom name, Average cost, Last Receipt Cost**, etc.

4.d All storeroom listed as filtering result.



Distributor	Manufacturer	Model	Last Price	Last Order Date	Order Unit	Organization	Site
708163			1,000.00	3/14/16	UNT	NPCT	NPCT1
709793			4,200,000.00	4/22/16	UNT	NPCT	NPCT1
MITSUI			1,000.00	2/22/16	EACH	NPCT	NPCT1

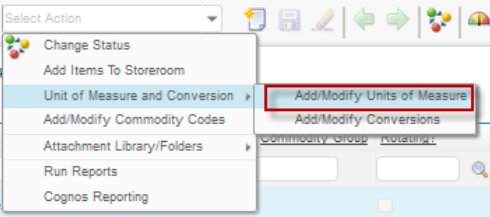
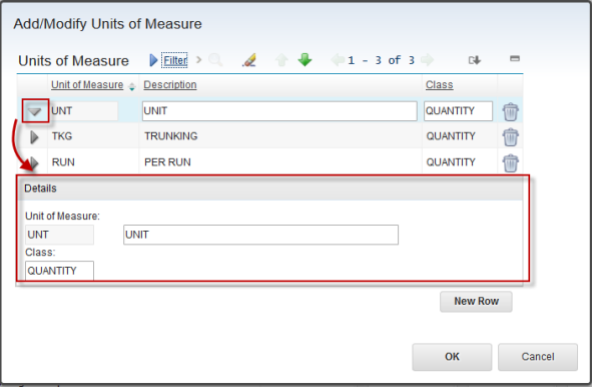
	5.a Distributors tab, to view all distributors related to this item 5.b Information about item's description 5.c Filter distributors by Distributor name, Manufacturer, Model, Last Price, etc. 5.d All distributors listed as filtering result.
6	 6.a Specification tab, to view specification toward the item 6.b Information about item's description 6.c Detail information about specifications for this item
7	 User can perform some action related to the item from Select Action menu (on top of the form). 7.a Change status action to change status this item 7.b Add Item To Storeroom, to register item to a storeroom, resulted in inventory record in Inventory application. 7.c Add/Modify Units of Measure, to view and modify units of measure master data 7.d Add/Modify Conversions, to view and modify conversions master data 7.e Add/Modify Commodity Code, to view and modify commodity groups and commodity codes master data 7.f View Status History, to view the change status history of the item. 7.g Add/Modify Image, to add/ modify image of the item. 7.h Item/Organization Details, to view/modify the status of item in specific organization. 7.i Duplicate item, to create new item by duplicating most value of the current item.

1.3.1 Unit of Measure & Conversion

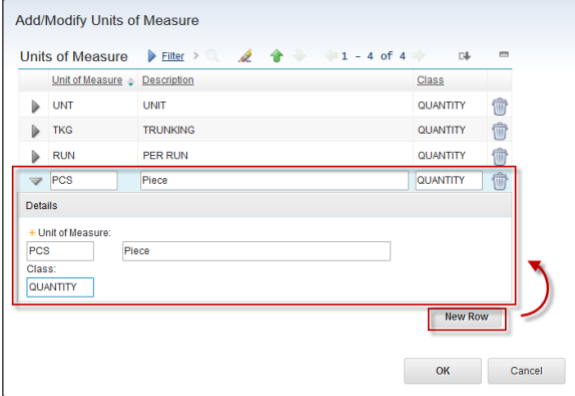
For some items, it is required to associate them with a unit of measure and conversion ratio. A unit of measure describes the increment in which you order, issue, or transfer an item or tool; for example, EACH, IN, or ROLL. A conversion ratio is a numeric value that relates one unit of measure to another. Unit of Measure and Conversion master data features can be accessed by *Procurement*.

When user add an item to a storeroom, user must supply a value for the issue unit using the units of measure that user have predefined. EMS will record the balance of the item in the storeroom based on the issue unit. Conversion takes place when user receive or transfer an item into a storeroom. If the Unit of Measure when user receive or transfer different from issue unit, conversion is the value correspond to the conversion between receiving or transfer unit and issue unit.

Below are steps to access, add, or modify Units of Measure data:

- 1 From **Item Master** application, on Select Action Menu, choose **Unit of Measure and Conversion > Add/Modify Units of Measure**. (It is not required to choose one of the item in Item Master application first)
- 
- 2 A popup window appears like below figure.
- 
- Unit of Measure** is the Unit of Measure Code. **Description** is the description of the unit of measure. **Class** is free text column to categorized the unit of measure, if necessary.
- Click the  button to show details of unit of measure. User can modify the description and class value here or in the grid. Click **OK** button to save the change, or **Cancel** button to cancel the change. Both button will close the popup window.

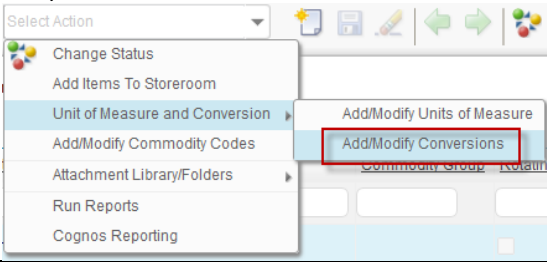
3



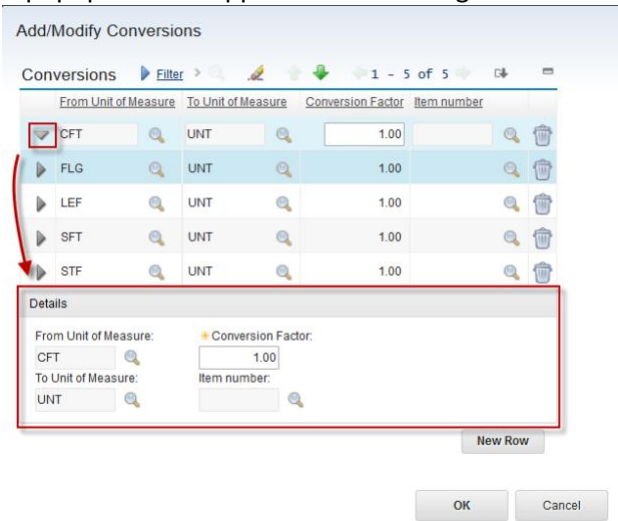
To add new unit of measure, click **New Row** button. A line is added to the grid. Provide the **Unit of Measure**, **Description**, and **Class** value as necessary. Then Click **OK** button to save or **Cancel** button to cancel, and the popup window will be closed. (User can add or modify another unit of measures before clicking OK button, to save all in one go)

Next, below are steps to access, add, or modify Conversions data:

1 From Item Master application, on Select Action Menu, choose **Unit of Measure and Conversion > Add/Modify Conversions**. (It is not required to choose one of the item in Item Master application first)



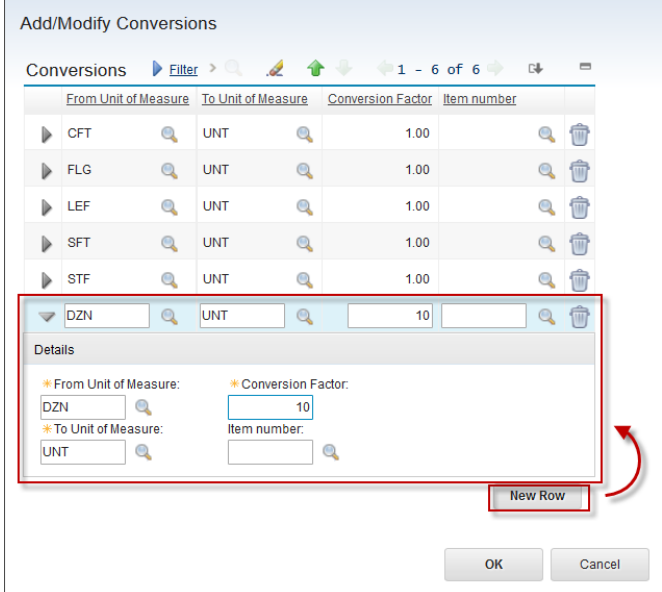
2 A popup window appears like below figure.



From Unit of Measure is the unit of measure that will be converted. **From Unit of Measure** is the unit of measure that will be converted to. **Conversion Factor** is the conversion value. **Item Number** is provided if the conversion is for specific item, otherwise, leave it blank.

Click the  button to show details of the conversion. User can modify the conversion value here or in the grid. Click **OK** button to save the change, or Cancel button to cancel the change. Both button will close the popup window.

3



To add new conversion, click **New Row** button. A line is added to the grid. Provide the **From Unit of Measure**, **To Unit of Measure**, **Conversion Factor**, and **Item number** value as necessary. Then Click **OK** button to save or **Cancel** button to cancel, and the popup window will be closed. (User can add another or modify another conversion before clicking OK button, to save all in one go)

As additional note, **Unit of Measure** and **Conversion** master data features can be accessed too from **Service Items** application, **Tools** application, **Purchase Requisition** application, and **Purchase Order** application.

1.3.2 Commodity Codes

There might be times when user need to manage items by their commodity. In addition, user might have the need to search for items by commodity or restrict purchases to specified commodities. Furthermore, commodity managers are sometimes tasked with reducing overall spending based on a commodity, and to analyze vendor utilization by commodity.

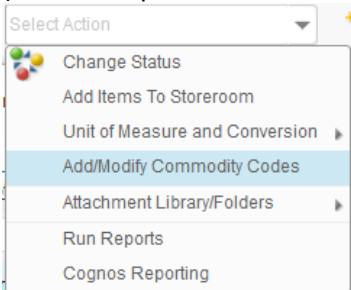
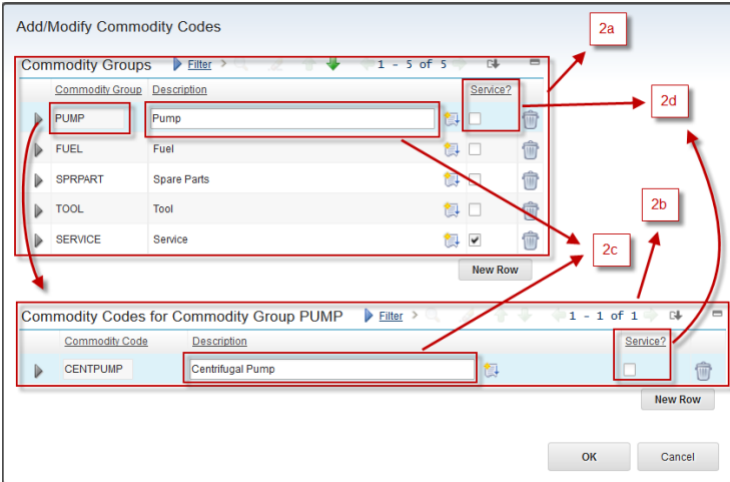
EMS supports two levels of commodities:

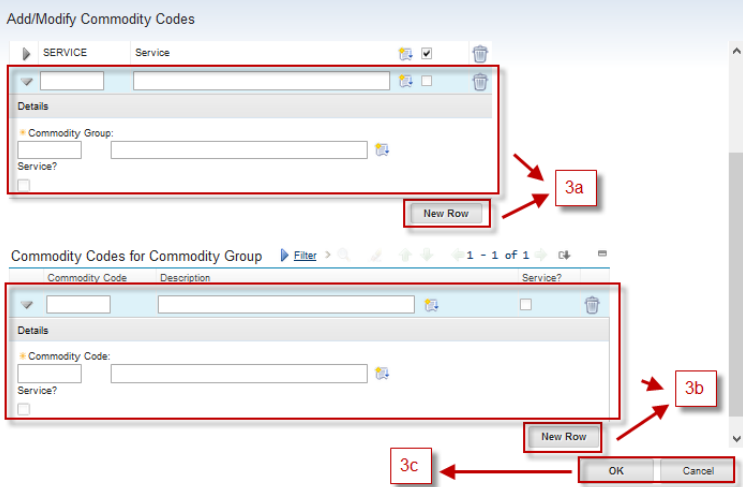
- **Commodity Group**, which is the broad classification of an item.
- **Commodity Code**, which is a second level of classification defined with the commodity group

By allowing user to associate an item to a commodity group and commodity code, user are able to analyze spending by transaction or by vendor based on the commodity group and code. Commodity Codes master data feature can be accessed by *Engineering Manager* and *Store Man* groups.

In addition to **Item Master** application, commodity codes master data feature can be accessed from **Service Item** application and **Tools** application too. Commodity group and commodity codes of **Item/Tool** type and **Service** type item must be different. A checkbox is available in each commodity groups and commodity codes, indicating whether it is used for **Service** type item, or **Item/Tool** type item.

Below are steps to access, add, or modify Commodity Group and Codes data:

- 1 From **Item Master** application, on Select Action Menu, choose **Add/Modify Commodity Codes**. (It is not required to choose one of the item in Item Master application first)
- 
- 2 A popup window will appears like below figure.
- 
- 3.a A table listing all commodity groups available.
 - 3.b A table listing all commodity codes of selected commodity group in table 2a.
 - 3.c To modify the description of a commodity group or code if necessary, via this fields.
 - 3.d To identify whether a commodity group or code is used for **Service** type or **Item/Tool** type item.

3


3.a To add new commodity group, click New Button in Commodity Group table, then provide the **Commodity Group**, **Description**, and **Service** checkbox value.

3.b To add new commodity code of selected commodity group, click New Button in Commodity Code table, then provide the **Commodity Code**, **Description**, and **Service** checkbox value.

3.c Click **OK** button to execute all changes, or **Cancel** to discard all changes.

1.3.3 Registering New Item

When user want to registering new **Item** type item master, user can create the item master from **Item Master** application. Created item needs to be integrated to Oracle GFS, because the procurement process will involve Oracle GFS system. After integration occurred and succeed, the item can be received or used in maintenance work. **Item Master** creation can be performed by *Engineering Manager*.

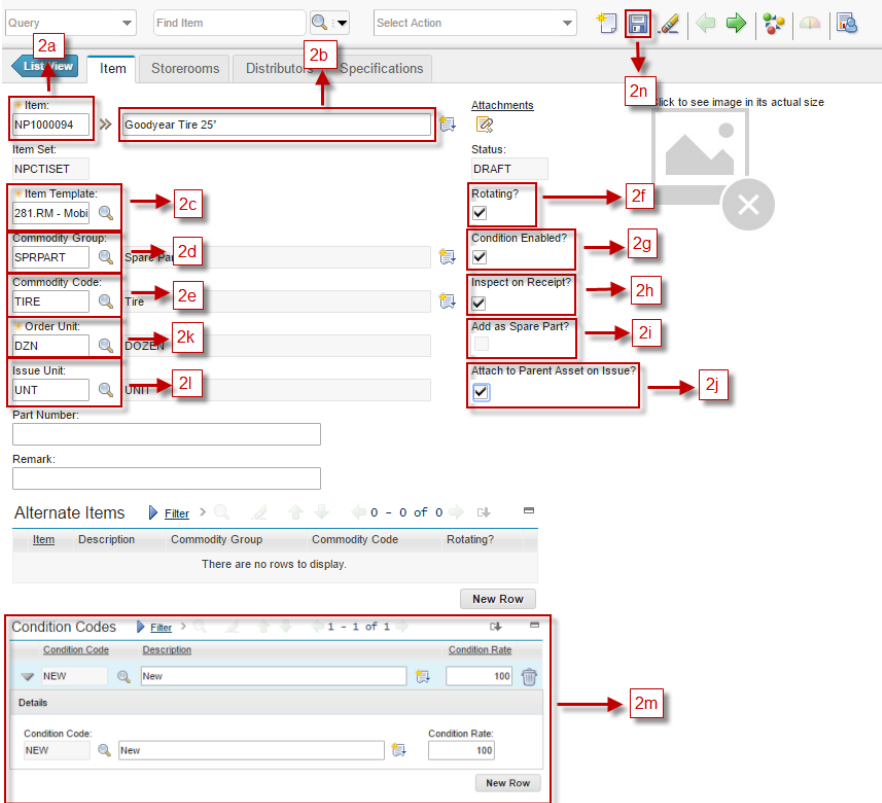
A point consideration about **Item** type item master is **Condition Enabled**. When an item is set into **Condition Enabled**, it means that user must define the condition of each item, whether it is new, used, reconditioned, etc. Condition of each item will affect the cost of the item when it is used.

In EMS, there are 2 conditions that is implemented, which are:

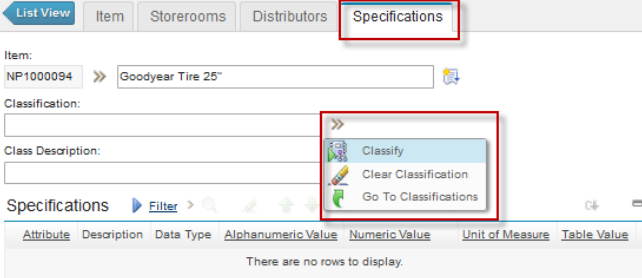
- **NEW** : Standard condition of the item.
- **RECOND** : Indicates reconditioning has been done to the item. The cost of this item will depends on the reconditioning work cost.

In EMS, **Condition Enabled** should be applied on **Rotating** item only, remembering that reconditioning work can be performed toward **Rotating** item only. When **Rotating** item have possibility of reconditioning against, it should be **Condition Enabled**. As additional notes, **Rotating** and **Condition Enable** can only be defined before the item is available in the system, so consider it carefully.

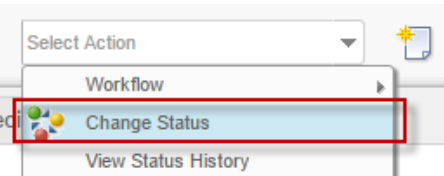
Below are steps for item master registration in **Item Master** application:

1	From Item Master application, click  button to add new item. User will be brought to Item tab with most field is empty.
2	 2.a Item Code number automatically filled by system 2.b Description of the item 2.c Information about Item Template of the item 2.d Information about Commodity Group of the item 2.e Information about Commodity Code of the Item 2.f Rotating check box, check if the item is rotating, uncheck it for non-rotating. 2.g Condition Enable check box, check if condition is enabled for the item, uncheck if not. 2.h Inspect on Receipt, check if the item need to go through inspection process upon receiving. 2.i Add as Spare Part, check if the item wants to be attached to asset as spare part automatically upon issued (this checkbox enable for non-rotating item only.) 2.j Attach to Parent Asset on Issue, check if the item wants to be attached to asset as subassembly automatically upon issued (this checkbox enable for rotating item only.) 2.k Information about Order Unit of the item 2.l Information about Issue Unit of the item 2.m Provide Condition codes that apply in the item, if Condition Enabled is checked. It is required to provide at least 1 Condition Code with condition rate 100, if the item is condition enabled. 2.n Click Save button to save.

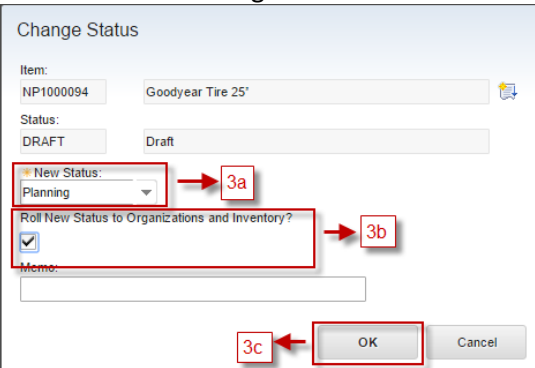
To add specification, go to **Specification** tab, and choose a classification from **Classify** feature. Please refer to below figure.



- 3 Next, user should change status this item from **DRAFT** to **PLANNING**, to initiate the integration process. Choose **Select Action – Change Status**



Show windows “change status”.

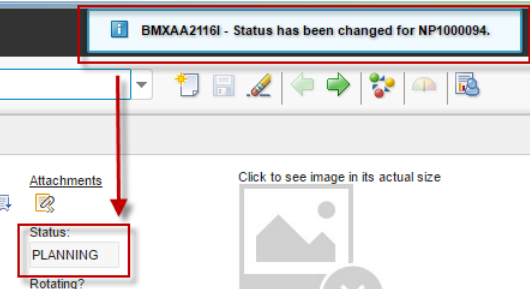


3.a **New Status**, filled status **Planning**

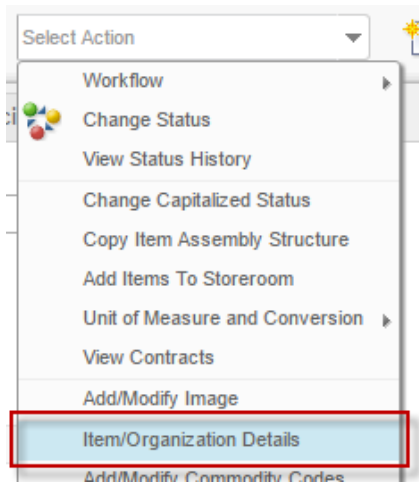
3.b **Roll New Status to Organizations and Inventory?**, check this box.

3.c Click **OK**.

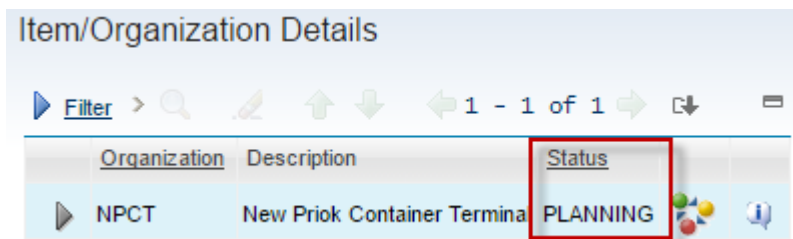
In the top page have information item changed and in field status is changed into **PLANNING**.



- 4 Check whether status of related item follows the status of the item master by **Select Action – Item/Organization Details**.

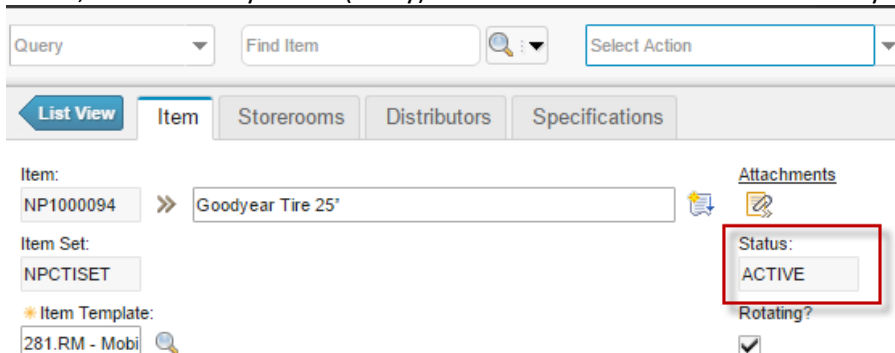


Show popup window “Item/Organization Details” and the status of the item in organization COMP is **PLANNING**.



If the status is still **DRAFT**, it means when user change the item master status into **PLANNING**, user forget to check the **Roll New Status to Organizations and Inventory?**. To change the status into **PLANNING** toward this Item/Organization Details, click the  button next to the record, and change the status accordingly.

- 5 If the integration is finished, system will change the item master status, item/organization details status, and Inventory status (if any) of the item into **ACTIVE** automatically.

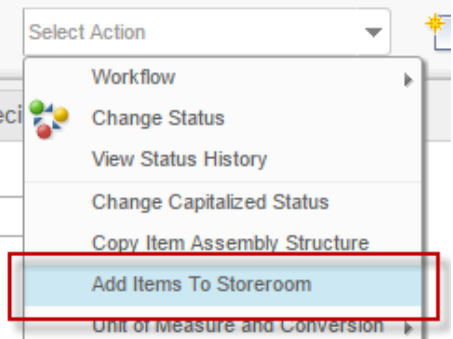
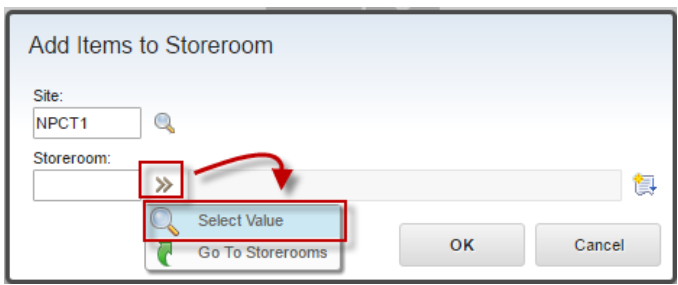


1.3.4 Add Item to Storeroom

After an item master is registered in **Item Master** application, the item should be registered in storeroom, through **Add Item to Storeroom** action. This process will create a record containing item data against chosen storeroom in **Inventory** application. **Inventory** application will be discussed more detail in other segment of this document. **Add Item to Storeroom** feature can be accessed by **Engineering Manager**.

By registering item in storeroom, reorder configuration can be configured toward those items in chosen storeroom via **Inventory** application. In this implementation, items should be procured based on the reorder configuration only. Therefore, **Add Item to Storeroom** must be performed on each new item master, to be able to be purchased.

Below are steps to perform **Add Item to Storeroom** action (reorder configuration will be discussed in **Inventory** application segment):

1	From Item Master application, choose one of the item which want to be added to storeroom. 
2	Detail of selected item is showed. Choose Select Action – Add Items to Storeroom 
3	Show popup window “Add Items to Storeroom”. Choose one storeroom available. User can be choose  and the Select Value. 

Show listed storeroom, choose one

Select Value

Filter > 1 - 2 of 2

Location	Description	Type	Site
		=STOREROO	
NPCT1-ENG-STORE	Engineering Storeroom	STOREROOM	NPCT1
NPCT1-IT-STORE	IT Storeroom	STOREROOM	NPCT1

Cancel

Storeroom filled and then click “OK”

Add Items to Storeroom

Site: NPCT1

Storeroom: NPCT1-ENG-: Engineering Storeroom

OK Cancel

Show another windows “Add Items to Storeroom”. Don’t change anything in this window, just click **OK** button.

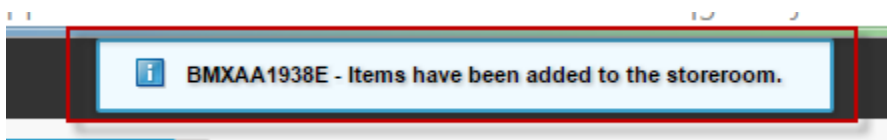
Add Items to Storeroom

Storeroom Information Filter > 1 - 1 of 1

Item	Issue Cost Type	Unit Cost	Default Bin	Current Balance	Lot	Issue Unit	Order Unit	Site
NP1000094	AVERAGE	0.00				UNT	DZN	NPCT1

OK Cancel

In the top page have information “Items have been added to the storeroom”



and in the **Storerooms** tab, added storeroom information appears.

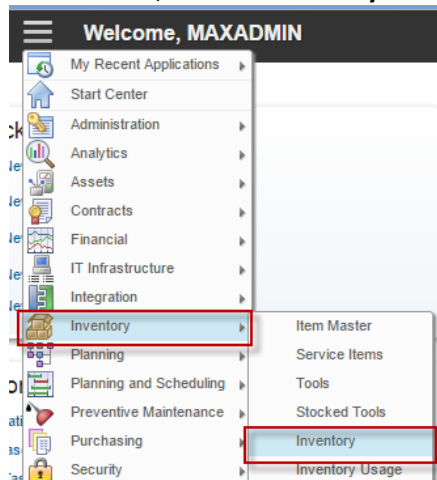
List View Item **Storerooms** Distributors Specifications

Item: NP1000094 Goodyear Tire 25 Item Set: NPCTISET

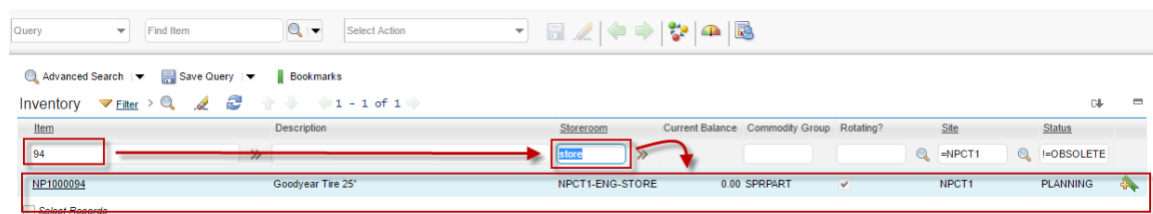
Storeroom Information Filter > 1 - 1 of 1

Storeroom	Average Cost	W Average Cost	W Average Cost 2	Last Receipt Cost	Current Balance	W Initial Balance	Default Bin	Status	Site
NPCT1-ENG-:	0.00	0.00	0.00	0.00	0.00	0.00		PLANNING	NPCT1

- 4 After item is added to storerooms, user can see in **Inventory** application (this application will be explained later) that a record is added, as a result of this “Add Item to Storeroom” process. From **Go To** menu, choose **Inventory – Inventory**



Perform filtering to look for the item and corresponding storeroom.



1.3.5 Change Status of Item

When user wants to deactivate, reactivate, or initiate integration process toward an item, user can do that by changing the status of item. Status changing can be performed using **Change Status** action. Not all item can be changed into another status via **Change Status** action at will. Every status required specific conditions of the item, to be able to be changed into, with details as follows:

- **DRAFT** : Initial status of new created Item. **Change Status** action can't change the item status into **DRAFT**.
- **PLANNING** : Only item with **DRAFT** Status can be changed into **PLANNING**, to initiate the integration process.
- **PENDOBS** : Only item with **ACTIVE** status can be changed into **PENDOBS**, when user plans to deactivate the item.
- **ACTIVE** : Only item with **PENDOBS** status can be changed into, to reactivate the item.
- **OBSOLETE** : Only item with **PENDOBS** status can be changed into, to deactivate the item.

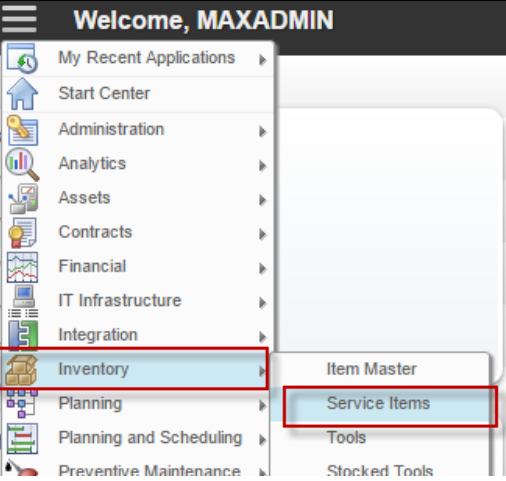
Change Status action details can refers to (1.3.3 Registering New Item) segment. **Change Status** feature can be accessed by *Engineering Manager*.

1.4 Service Items Application

Service Items application used to add or maintain item master data with the type of **Service**. Frequent purchased service should be catalogued in EMS, to track the purchases and usages history of the service easily. Because it is a service, it won't be stocked in storerooms.

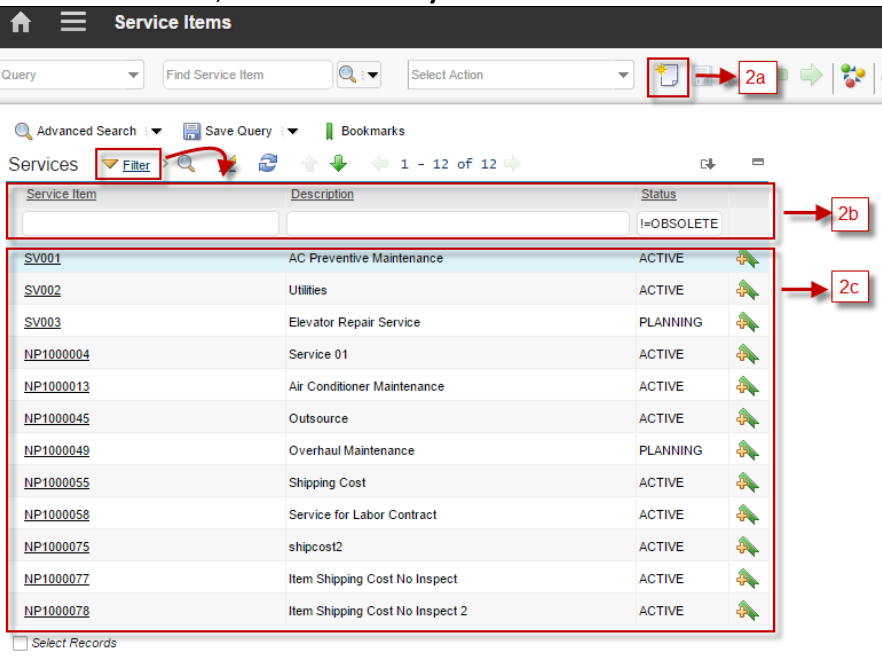
Followings are details about **Service Item** applications accessibility:

1



From **Go To** menu, choose **Inventory – Service Items**

2



Header Bar will give information that user is in the **Service Items** application.

2.a **New Service Item**, to add new service item

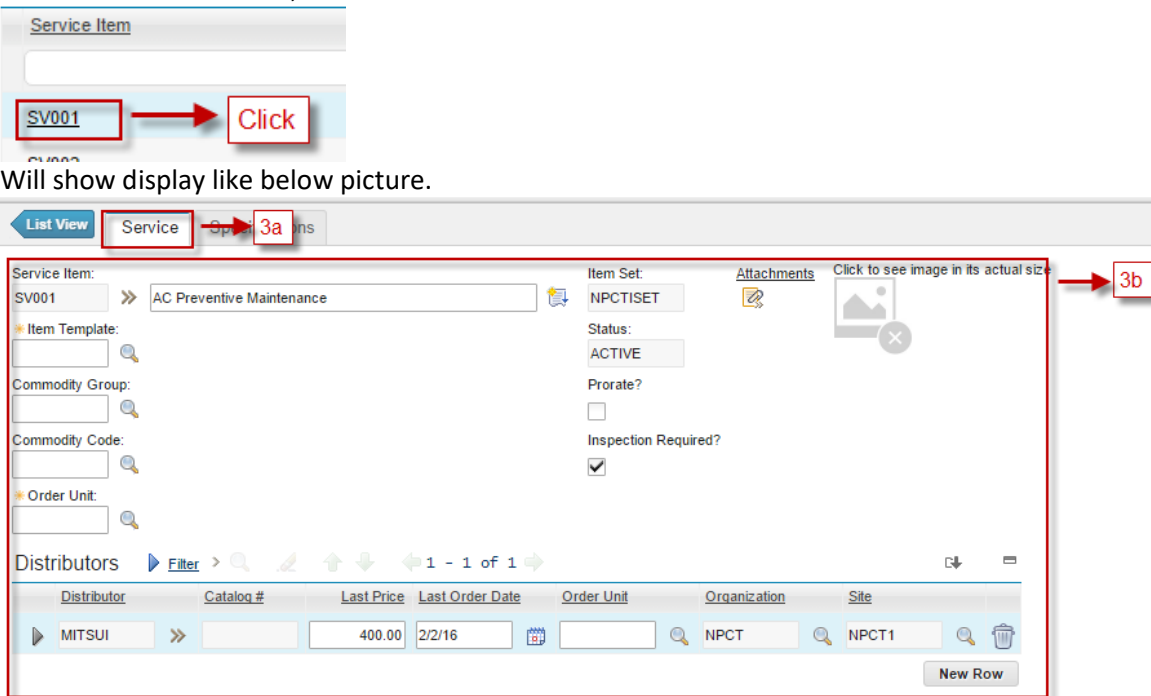
2.b **Filter** service items by **Service Item** code, **Description**, and **Status**

2.c Listed all service items, based on filtering result.

Agile – Proprietary/Confidential

23

3 Choose one service item, to show details of service item



Service Item

SV001 Click

Will show display like below picture.

List View Service Specifications

Service Item: SV001 >> AC Preventive Maintenance

Item Set: NPCTISET

Status: ACTIVE

Prorate? ☐

Inspection Required? ☒

Attachments Click to see image in its actual size

Distributors

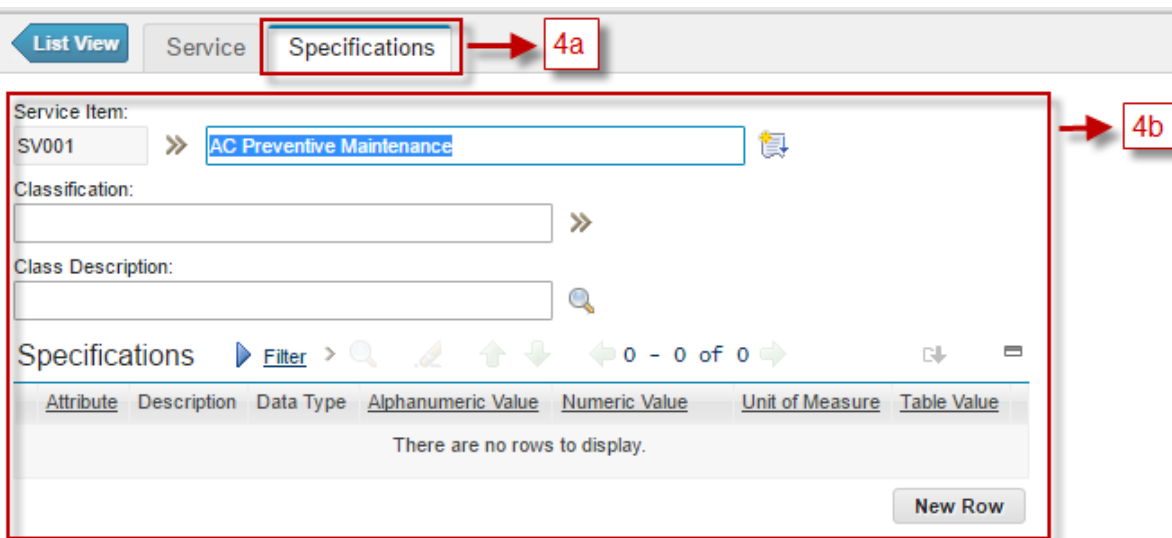
Distributor	Catalog #	Last Price	Last Order Date	Order Unit	Organization	Site
MITSUI		400.00	2/2/16		NPCT	NPCT1

New Row

3.a **Service** tab, to view and update details of service.

3.b Detail information of the item, such as item set, item template, commodity group and codes, Order unit, prorate, etc.

4 Choose **Specifications** tab, besides **Service** tab



List View Service Specifications

Service Item: SV001 >> AC Preventive Maintenance

Classification:

Class Description:

Specifications

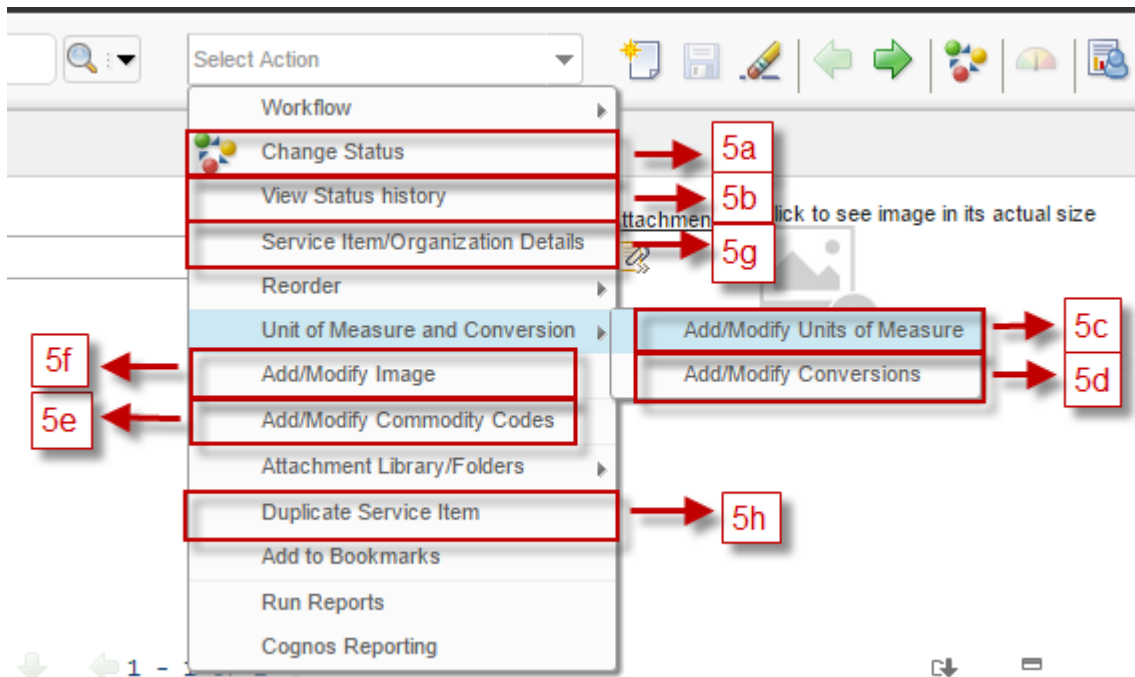
Attribute	Description	Data Type	Alphanumeric Value	Numeric Value	Unit of Measure	Table Value
There are no rows to display.						

New Row

4.a **Specifications** tab, to view and update specifications of service.

4.b Detail information about specifications for this service.

5 User can perform some action related to the service from Select Action menu (on top of the form).



5.a **Change Status** action, to change the status of service.

5.b **View Status History**, to view the change status history of the service.

5.c **Add/Modify Units of Measure**, to view and modify units of measure master data

5.d **Add/Modify Conversions**, to view and modify conversions master data

5.e **Add/Modify Commodity Code**, to view and modify commodity groups and commodity codes master data

5.f **Add/Modify Image**, to add/ modify image of the service.

5.g **Service Item/Organization Details**, to view/modify the status of service in specific organization.

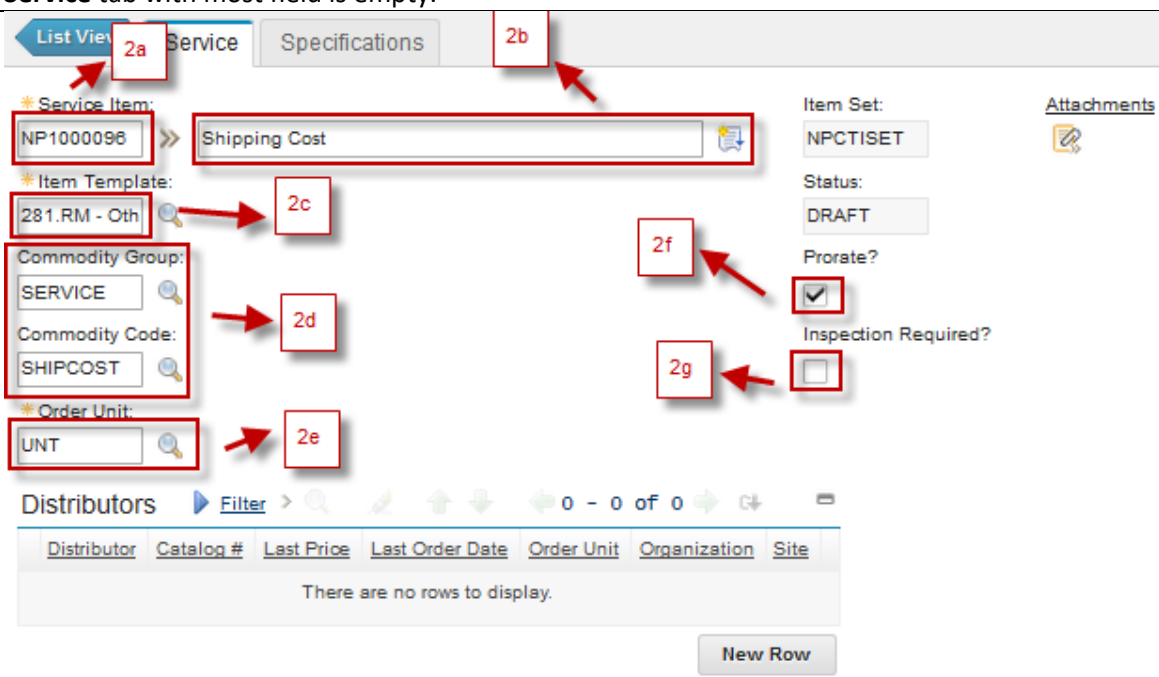
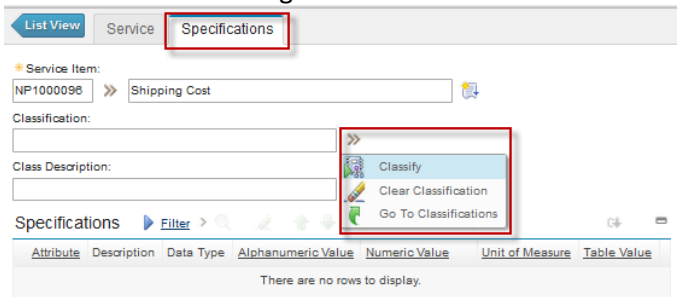
5.h **Duplicate Service Item**, to create new service by duplicating most value of the current service.

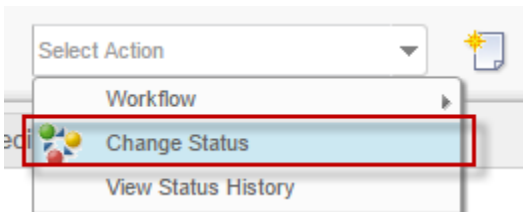
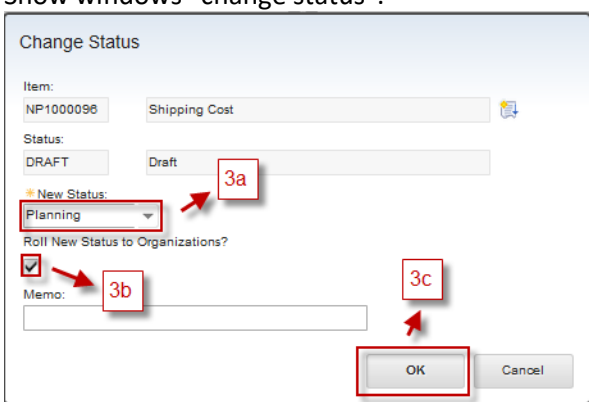
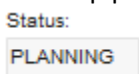
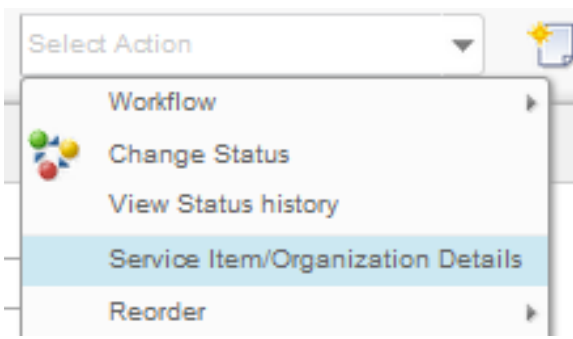
1.4.1 Registering New Service

When user want to registering new **Service** type item master, user can create the item master from **Service Item** application. Created service needs to be integrated to Oracle GFS, because the procurement process will involve Oracle GFS system. After integration occurred and succeed, the service can be received or used in maintenance work. **Service** item creation can be performed by *Engineering Manager*.

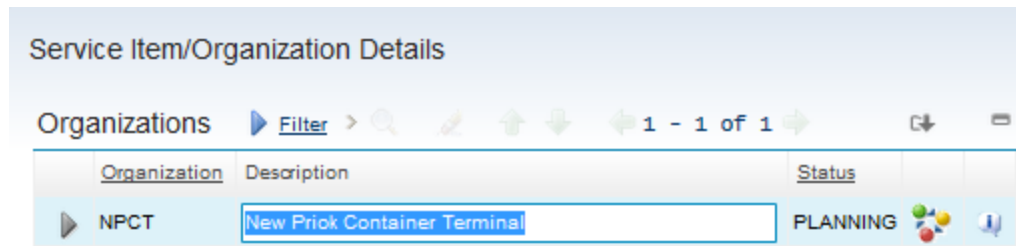
A point consideration about Service Item is **Prorate** feature. If a service is treated as **Prorate** service, when it is purchased, the value will be prorated to all items purchased in the same PO, based on each item cost. In this implementation, shipping cost service is an example that should be treated as **Prorate** service.

Below are steps for service item registration in **Service Item** application:

1	From Service Item application, click  button to add new service. User will be brought to Service tab with most field is empty.
2	 2.a Service Item Code number automatically filled by system 2.b Description of the service 2.c Information about Item Template of the service 2.d Information about Commodity Group and Commodity Code of the service 2.e Information about Order Unit of the service 2.f Prorate check box, check if the service is prorate, and uncheck it for non-prorate. 2.g Inspect on Receipt, check if the service need to go through inspection process upon receiving. To add specification, go to Specification tab, and choose a classification from Classify feature. Please refer to below figure. 

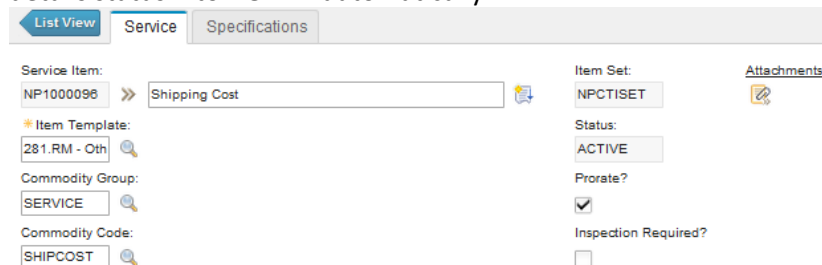
	Click Save  to save.
3	Next, user should change status this service from DRAFT to PLANNING, to initiate the integration process. Choose Select Action – Change Status  Show windows “change status”.  3.d New Status, filled status Planning 3.e Roll New Status to Organizations?, check this box. 3.f Click OK. In the top page have information service changed and in field status is changed into PLANNING. 
4	Check whether status of related service follows the status of the service by Select Action – Service Item/Organization Details. 

Show popup window “Service Item/Organization Details” and the status of the service in organization COMP is **PLANNING**.



If the status is still **DRAFT**, it means when user change the service status into **PLANNING**, user forget to check the **Roll New Status to Organizations?**. To change the status into **PLANNING** toward this Service Item/Organization Details, click the  button next to the record, and change the status accordingly.

5 If the integration is finished, system will change the service status and service item/organization details status into **ACTIVE** automatically.



1.4.2 Change Status of Service

Same as **Item Master** application, **Change Status** action can be used to deactivate, reactivate, or initiate integration process toward a service. Not all service can be changed into another status via **Change Status** action at will. Every status required specific conditions of the service, to be able to be changed into, with details as follows:

- **DRAFT** : Initial status of new created service. **Change Status** action can't change the service status into **DRAFT**.
- **PLANNING** : Only service with **DRAFT** Status can be changed into **PLANNING**, to initiate the integration process.
- **PENDOBS** : Only service with **ACTIVE** status can be changed into **PENDOBS**, when user plans to deactivate the service.
- **ACTIVE** : Only service with **PENDOBS** status can be changed into, to reactivate the service.
- **OBSOLETE** : Only service with **PENDOBS** status can be changed into, to deactivate the service.

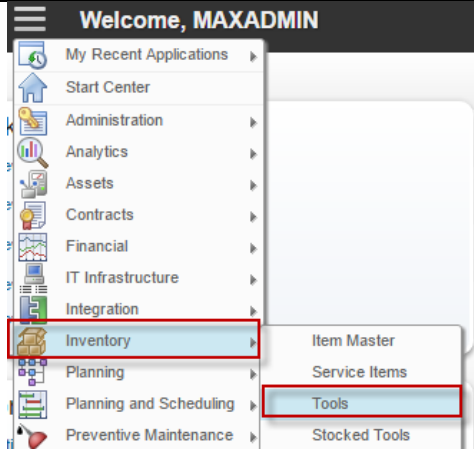
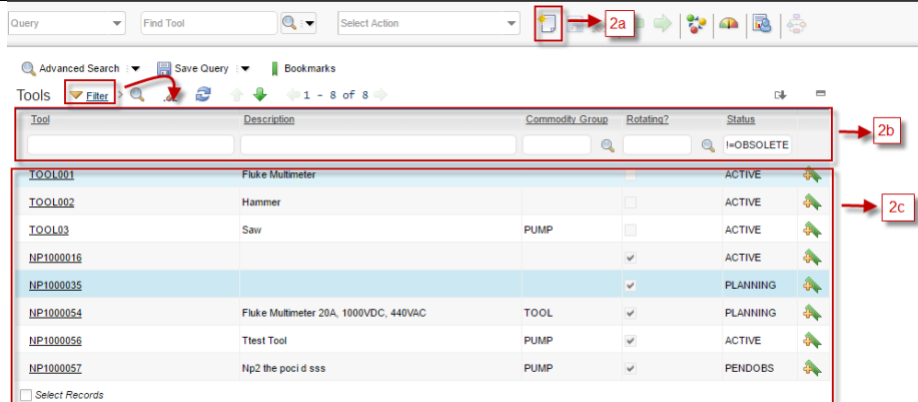
Change Status action details can refer to (1.4.1 Registering New Service) segment. **Change Status** feature can be accessed by *Engineering Manager*.

1.5 Tools Application

Tools application used to add or maintain item master data with the type of **Tool**. As stated before, the notable differences between **Items** and **Tools** are the items consumed, while tool is not. Tool can be stored in Storeroom too. Tool usage in maintenance work won't be charged.

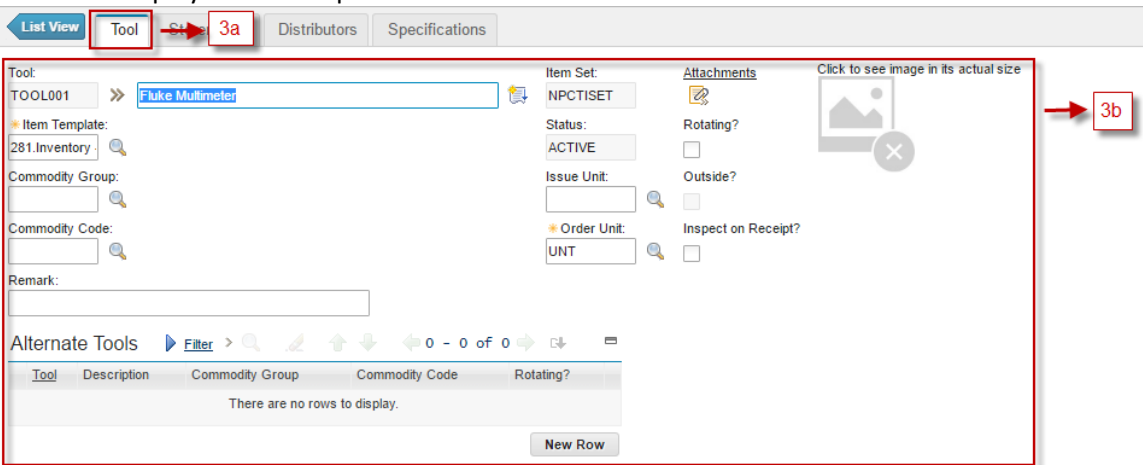
Tool type item can be treated as asset by setting them into **Rotating**, which means each tool will have an asset record on their own in **Asset** application. Therefore, a maintenance work can be performed toward those tools too. Besides, this **Rotating** feature can help user to track usage history of specific tools.

Followings are details about **Tools** applications accessibility:

1	 From Go To menu, choose Inventory - Tools
2	 2.a New Tool, to add new tool 2.b Filters tools by Tool Code, Description, Commodity Group, Rotating and Status 2.c Listed all tools based on filtering result.
3	Choose one tool, to show details of tool



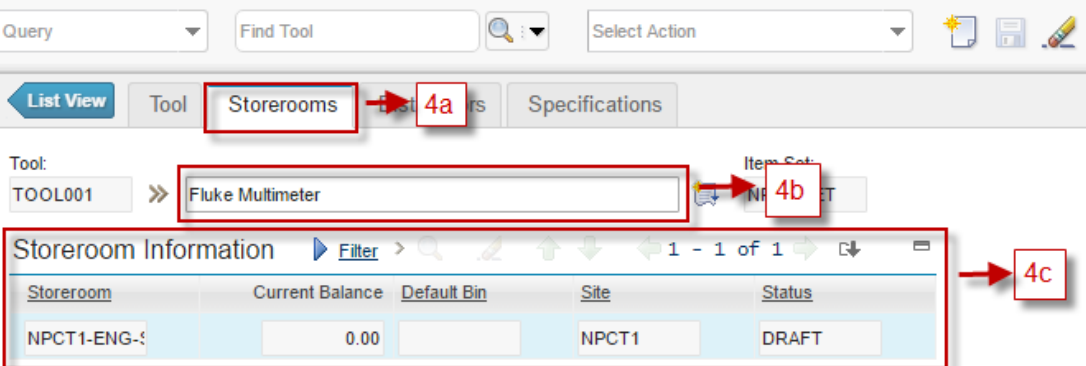
Will show display like below picture.



3.a **Tool** tab, to view and update details of tool

3.b Detail information about tool, such as Tool Code, Description, Item Template, Commodity, Issue and Order Unit, Rotating, etc.

4 Choose **Storerooms** tab besides the **Tool** tab

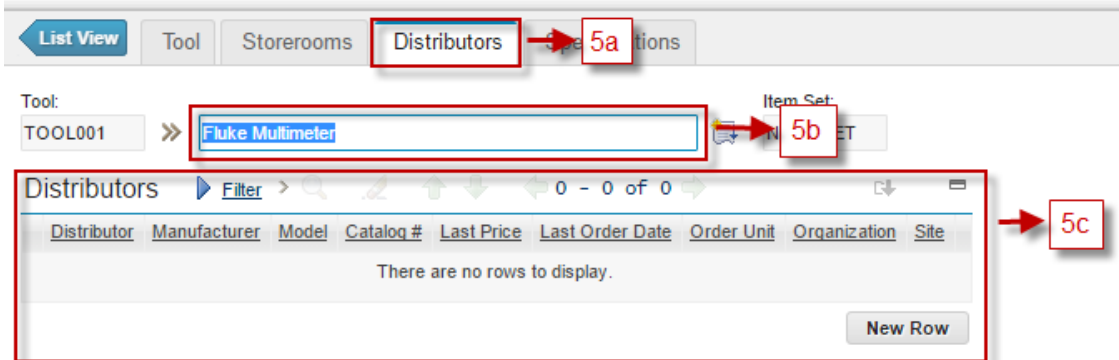


4.a **Storerooms** tab, to view all storerooms related to tool

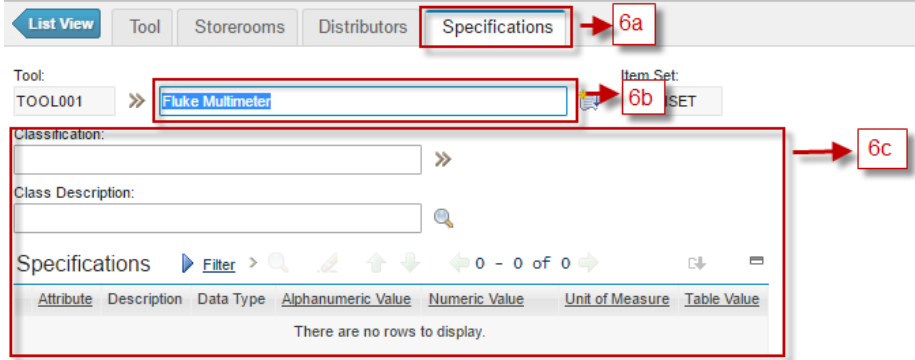
4.b Information about tool's description

4.c Listed all storerooms related to tool

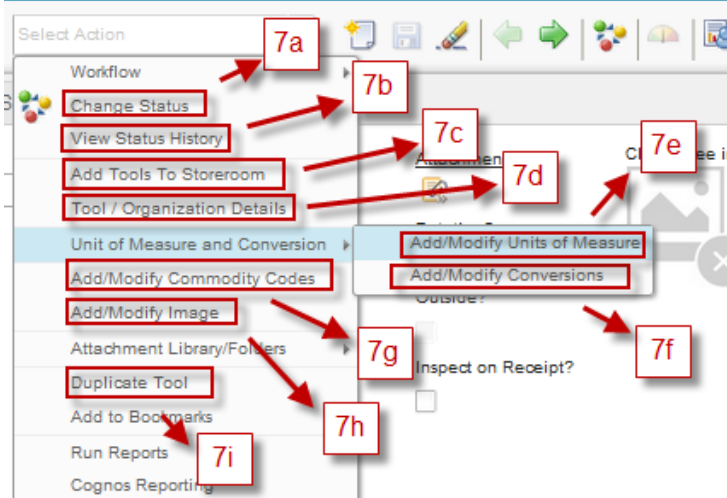
5 Choose **Distributors** tab besides **Storerooms** tab



5.a **Distributors** tab, to view all distributors related to tool.
5.b Information about tool's description
5.c Listed all distributors related to tool.



6 Choose **Specifications** tab besides **Distributors** tab
6.a **Specifications** tab: to view and update specifications about related tool
6.b Information about tool's description
6.c Information about details specification of tool



7 User can perform some action related to the tool from Select Action menu (on top of the form).

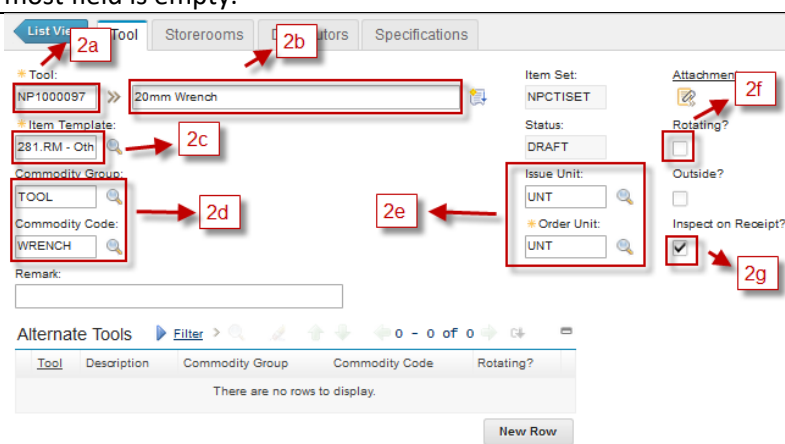
7.a Change Status : action for change status tool
7.b View Status History , to view the change status history of the tool.
7.c Add Tools To Storeroom : action for add tools to storeroom
7.d Tool/Organization Details , to view/modify the status of tool in specific organization.
7.e Add/Modify Units of Measure , to view and modify units of measure master data
7.f Add/Modify Conversions , to view and modify conversions master data
7.g Add/Modify Commodity Code , to view and modify commodity groups and commodity codes master data
7.h Add/Modify Image , to add/ modify image of the tool.
7.i Duplicate Tools , to create new tool by duplicating most value of the current tool.

1.5.1 Register New Tools

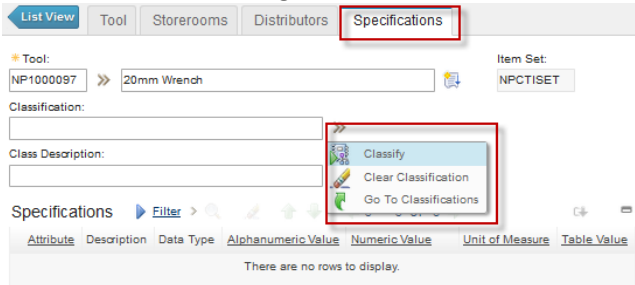
When user want to registering new **Tool** type item master, user can create the item master from **Tools** application. Created tool needs to be integrated to Oracle GFS, because the procurement process will involve Oracle GFS system. After integration occurred and succeed, the item can be received or used in maintenance work. **Tools** creation can be performed by *Engineering Manager*.

A tool can be set into **Rotating**, where it is better to be defined early, before it is purchased. Once it is available in the system, it cannot be changed anymore. So, consider it carefully.

Below are steps for tool registration in **Tool** application:

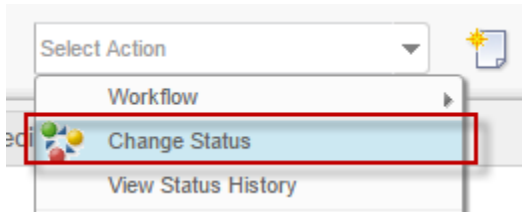
1	From Tools application, click  button to add new tool. User will be brought to Tool tab with most field is empty.
2	 2.a Tool Code number automatically filled by system 2.b Description of the tool 2.c Information about Item Template of the tool 2.d Information about Commodity Group and Commodity Code of the tool 2.e Information about Order Unit and Issue Unit of the tool 2.f Rotating check box, check if the tool is rotating, and uncheck it for non-rotating. 2.g Inspect on Receipt, check if the tool need to go through inspection process upon receiving.

To add specification, go to **Specification** tab, and choose a classification from **Classify** feature. Please refer to below figure.

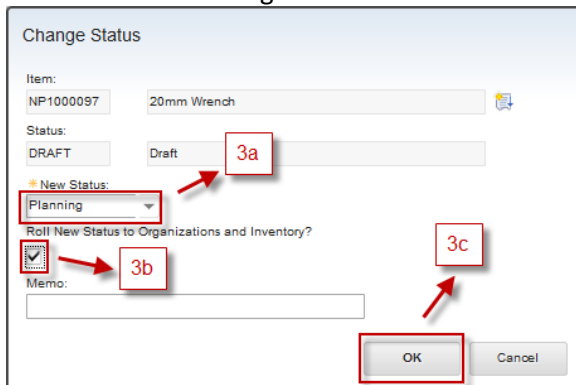


Click Save  to save.

- 3 Next, user should change status this service from **DRAFT** to **PLANNING**, to initiate the integration process. Choose **Select Action – Change Status**



Show windows “change status”.

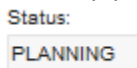


3.g **New Status**, filled status **Planning**

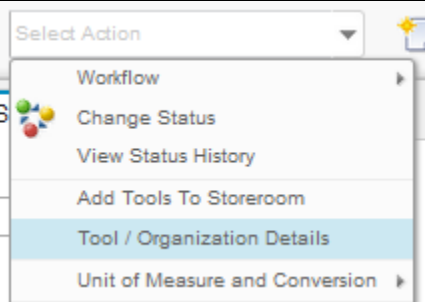
3.h **Roll New Status to Organizations and Inventory?**, check this box.

3.i Click **OK**.

In the top page have information tool changed and in field status is changed into **PLANNING**.



- 4 Check whether status of related tool follows the status of the tool master by **Select Action – Tool/Organization Details**.



Show popup window “Tool Item/Organization Details” and the status of the tool in organization COMP is **PLANNING**.

Tool/Organization Details

Organizations [Filter](#) >     1 - 1 of 1 

Organization	Description	Status	Tool Rate	
NPCT	New Prick Container Terminal	PLANNING	0.00	 

If the status is still **DRAFT**, it means when user change the tool status into **PLANNING**, user forget to check the **Roll New Status to Organizations and Inventory?**. To change the status into **PLANNING** toward this Tool/Organization Details, click the  button next to the record, and change the status accordingly.

- 5 If the integration is finished, system will change the tool status, tool/organization details status, and stocked tool status (if any) into **ACTIVE** automatically.

[List View](#) **Tool** [Storerooms](#) [Distributors](#) [Specifications](#)

Tool: NP1000097 >> 20mm Wrench 

* Item Template: 281.RM - Oth 

Commodity Group: TOOL 

Commodity Code: WRENCH 

Remark:

Item Set: NPCTISET

Status: **ACTIVE**

Issue Unit: UNT 

* Order Unit: UNT 

[Attachments](#) 

Rotating? ☐

Outside? ☐

Inspect on Receipt? ☒

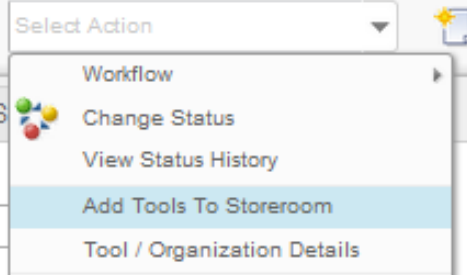
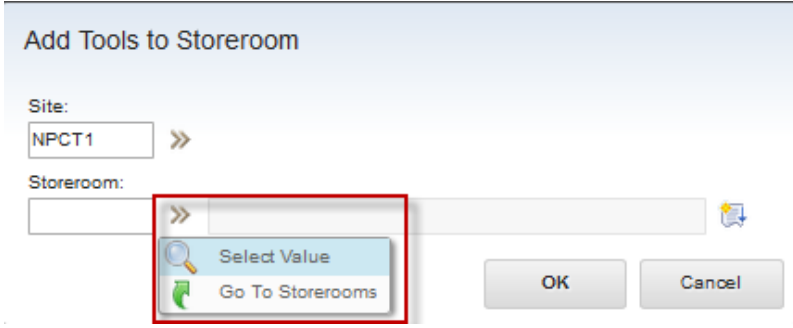
1.5.2 Add Tool to Storeroom

After tool is registered in **Tools** application, the tool can be registered in storeroom, through **Add Tool to Storeroom** action. This process will create a record containing tool data against chosen storeroom in **Stocked Tools** application. **Stocked Tools** application will be discussed more detail in other segment of this document. **Add Tool to Storeroom** feature can be accessed by *Engineering Manager*.

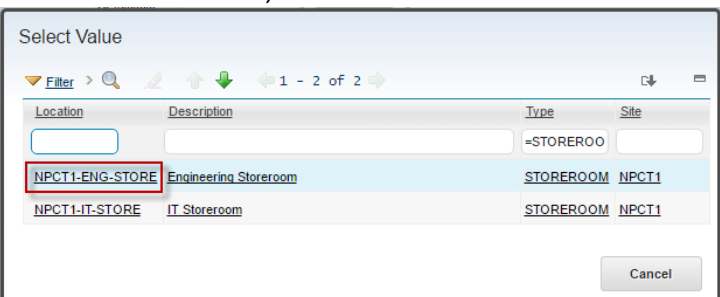
Different from items, there is no reorder feature for tools, because tools are not consumed. So, in this implementation, it is not necessary to perform this actions. Purchasing can be initiated manually toward tools. A record in **Stocked Tools** application will be added automatically once a receiving performed toward those tool in specific storeroom.

Below are steps to perform **Add Tool to Storeroom**:

- 1 From **Tool Master** application, choose one of the tool which want to be added to storeroom.

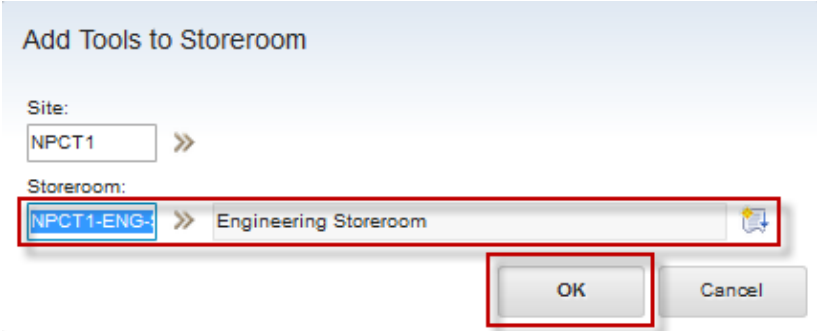
- 2 Detail of selected tool is showed. Choose **Select Action – Add Tools to Storeroom**

- 3 Show popup window “Add Tools to Storeroom”. Choose one storeroom available. User can be choose  and the **Select Value**.


Show listed storeroom, choose one



Location	Description	Type	Site
		=STOREROO	
NPCT1-ENG-STORE	Engineering Storeroom	STOREROOM	NPCT1
NPCT1-IT-STORE	IT Storeroom	STOREROOM	NPCT1

Storeroom filled and then click “OK”



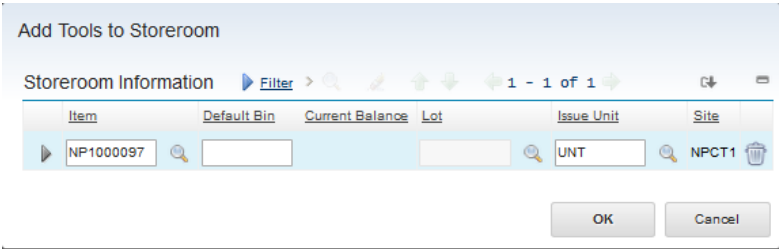
Add Tools to Storeroom

Site: NPCT1 >>

Storeroom: NPCT1-ENG- >> Engineering Storeroom

OK Cancel

Show another windows “Add Tools to Storeroom”. Don’t change anything in this window, just click **OK** button.



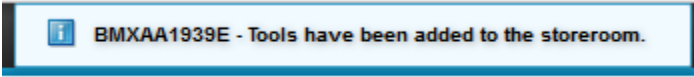
Add Tools to Storeroom

Storeroom Information Filter > 1 - 1 of 1

Item	Default Bin	Current Balance	Lot	Issue Unit	Site
NP1000097		0.00		UNT	NPCT1

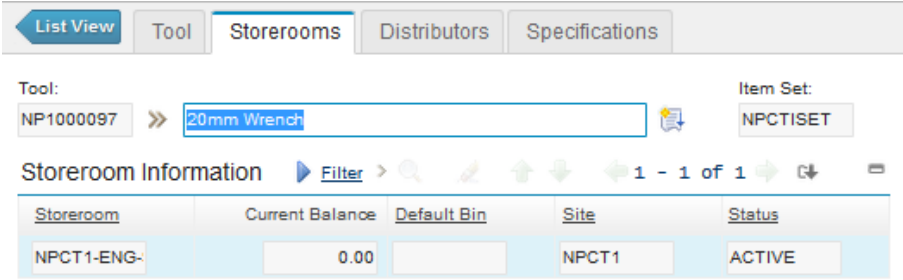
OK Cancel

In the top page have information “Items have been added to the storeroom”



BMXAA1939E - Tools have been added to the storeroom.

and in the **Storerooms** tab, added storeroom information appears.



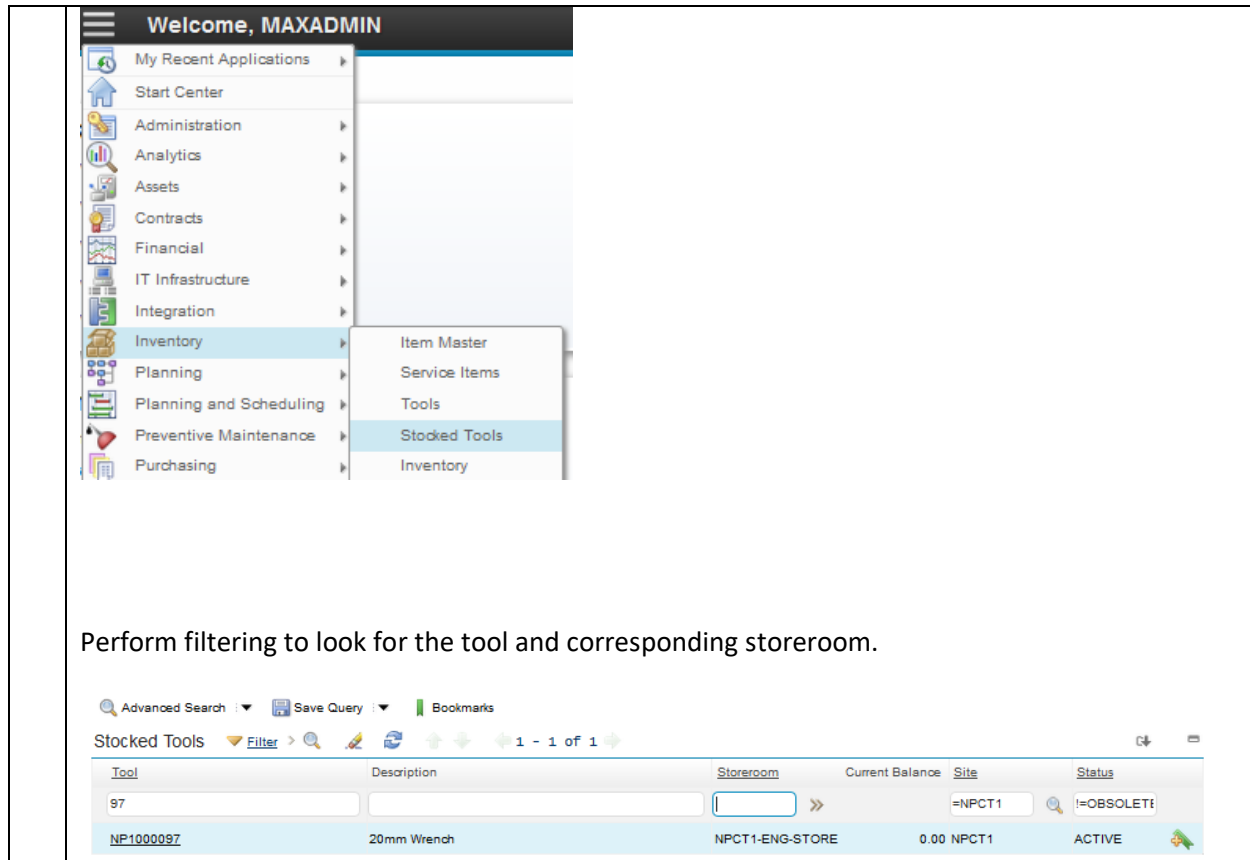
List View Tool Storerooms Distributors Specifications

Tool: NP1000097 >> 20mm Wrench Item Set: NPCT1SET

Storeroom Information Filter > 1 - 1 of 1

Storeroom	Current Balance	Default Bin	Site	Status
NPCT1-ENG-	0.00		NPCT1	ACTIVE

- 4 After tool is added to storerooms, user can see in **Stocked Tools** application (this application will be explained later) that a record is added, as a result of this “Add Tool to Storeroom” process. From **Go To** menu, choose **Inventory – Stocked Tools**



The screenshot shows the MAXADMIN application interface. The top navigation bar includes 'Welcome, MAXADMIN' and a list of application categories: My Recent Applications, Start Center, Administration, Analytics, Assets, Contracts, Financial, IT Infrastructure, Integration, Inventory, Planning, Planning and Scheduling, Preventive Maintenance, and Purchasing. The 'Inventory' category is expanded, showing sub-items: Item Master, Service Items, Tools, Stocked Tools, and Inventory. The 'Stocked Tools' sub-item is selected, leading to a table view.

Below the navigation bar, there is a search and filter section with 'Advanced Search', 'Save Query', and 'Bookmarks' buttons. The 'Stocked Tools' table is displayed with the following columns: Tool, Description, Storeroom, Current Balance, Site, and Status. The table contains one row of data:

Tool	Description	Storeroom	Current Balance	Site	Status
NP1000097	20mm Wrench	NPCT1-ENG-STORE	0.00 NPCT1	ACTIVE	

Perform filtering to look for the tool and corresponding storeroom.

1.5.3 Change Status of Tool

Same as **Item Master** and **Service Items** application, **Change Status** action can be used to deactivate, reactivate, or initiate integration process toward a tool. Not all tools can be changed into another status via **Change Status** action at will. Every status required specific conditions of the tool, to be able to be changed into, with details as follows:

- **DRAFT** : Initial status of new created tool. **Change Status** action can't change the tool status into **DRAFT**.
- **PLANNING** : Only tool with **DRAFT** Status can be changed into **PLANNING**, to initiate the integration process.
- **PENDOBS** : Only tool with **ACTIVE** status can be changed into **PENDOBS**, when user plans to deactivate the tool.
- **ACTIVE** : Only tool with **PENDOBS** status can be changed into, to reactivate the tool.
- **OBSOLETE** : Only tool with **PENDOBS** status can be changed into, to deactivate the tool.

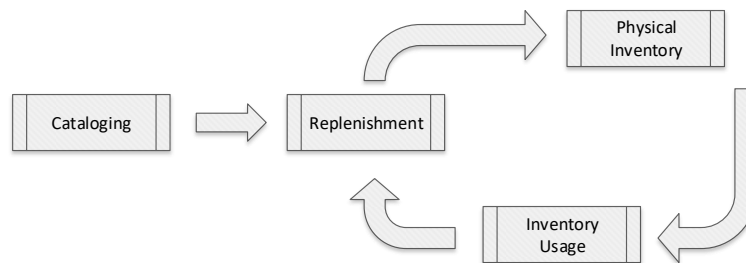
Change Status action details can refers to (1.5.1 Register New Tools) segment. **Change Status** feature can be accessed by *Engineering Manager*.

1.6 Exercise Phase

2 Inventory

2.1 Overview

Inventory is one of the central modules in Maximo. It functions in a dynamic relationship with Work Orders, Preventive Maintenance, Purchase Orders, Assets, Tools, and Companies applications. These other applications affect the quantity of items in inventory and identify where those items are needed or located, as well as who sells them.



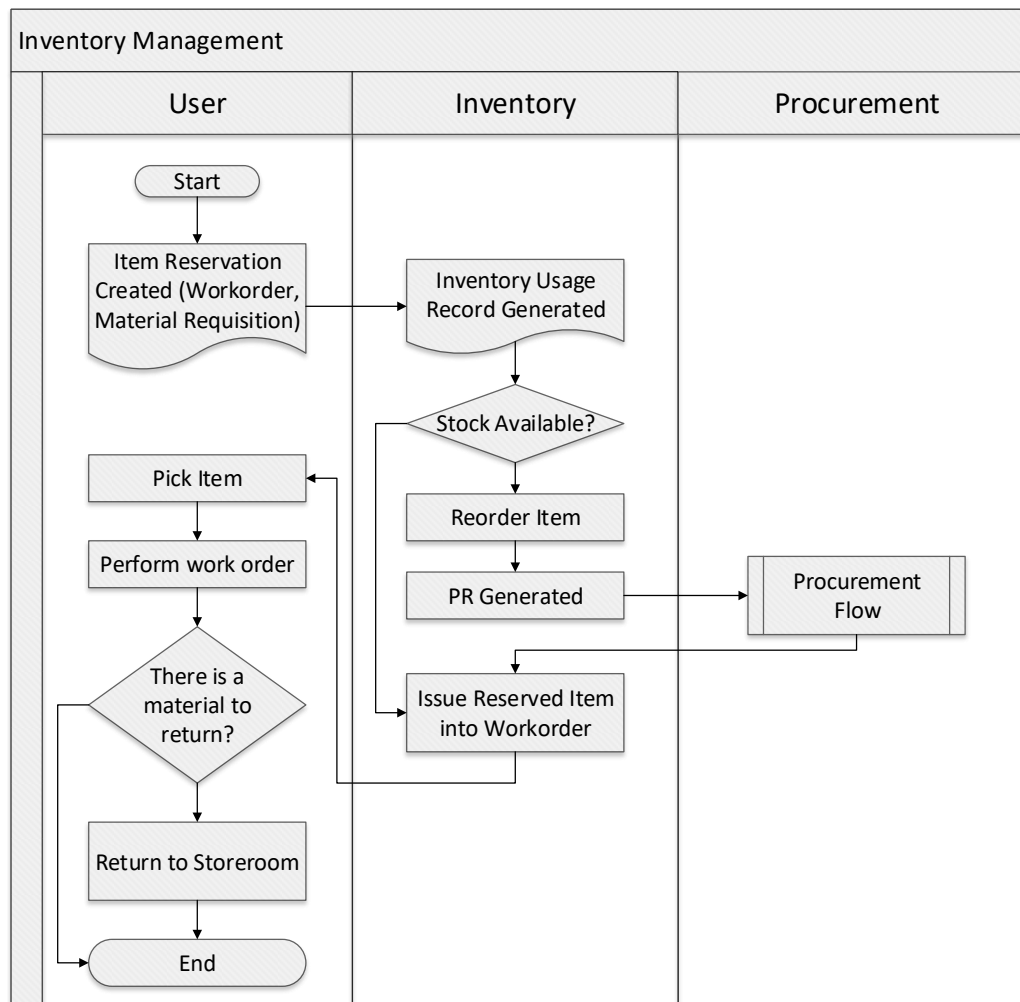
Managing inventory is an important part of maintaining any asset because it will help to handle items needed for maintenance activities. Above figure show how all process of inventory management related and managed to help COMP1 to keep track of following activities:

1. Items cataloguing (already discussed in **1. Item, Service, and Tool** segment).
2. Inventory Storage.
3. Inventory replenishment.
4. Inventory usage transaction (issue/return/adjustment).

As general, Inventory transaction cycle flow is as follows:

1. Inventory reservation is required to issue an item from storeroom. The sources of inventory reservation are:
 - Planned material from approved work order.
 - Approved material request for unplanned material that is required for a work order.Every inventory reservation will automatically create a usage document to issue an item from storeroom.
2. If the reserved item is sufficient, *Store Man* can issue that item. Else, procurement process should be initiated to provide the item. After the item is procured, *Store Man* can issue that item. Inventory issue should be recorded in usage document, automatically from inventory reservation (on step 1), or manually by creating new usage document.
3. After work order is completed and if there are any items that need to be returned to storeroom, technician should bring the item to *Store Man*. A usage document should be created to record the inventory return.

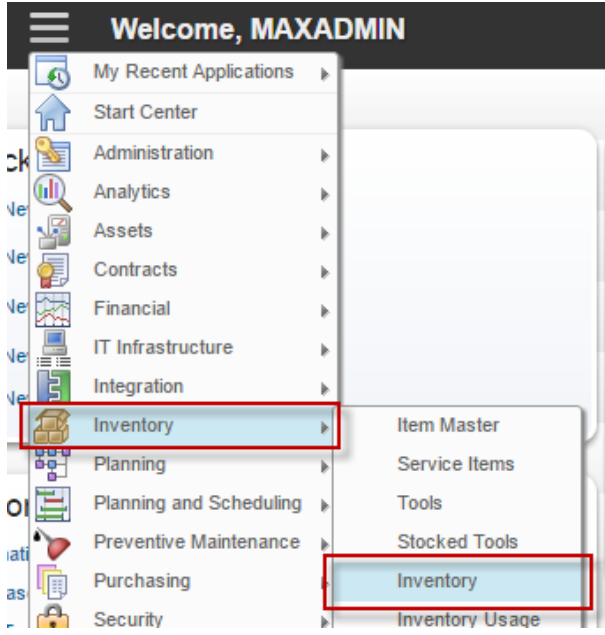
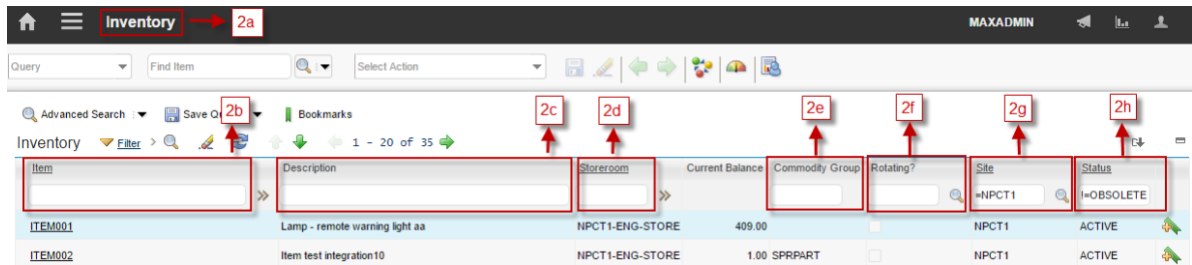
Below are detailed figure about inventory transaction cycle flow.



2.2 Inventory Application

Inventory is a listing of all items that are stored in storerooms (called *inventory data*). Using the **Inventory** application, user can track item balances down to the bin and lot level for a storeroom, as well as item costs per condition codes if Condition Codes applied on the items. Replenishment activity can be initiated from **Inventory** application through reorder configuration. Inventory data created when user perform **Add Item to Storeroom** action from **Item Master** application.

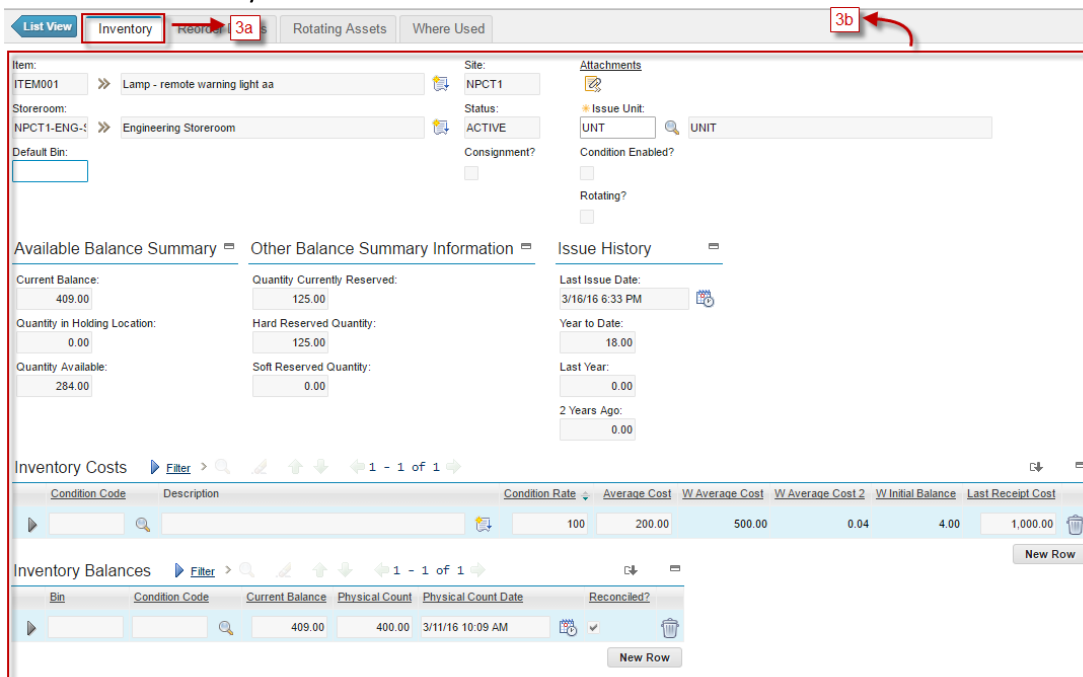
Followings are details about **Inventory** applications accessibility:

- 1 From **Go To** menu, choose **Inventory – Inventory**

- 2 Show all listed inventory data based on filtering result.
 

Item	Description	Storeroom	Current Balance	Commodity Group	Rotating?	Site	Status
ITEM001	Lamp - remote warning light aa	NPCT1-ENG-STORE	409.00			NPCT1	ACTIVE
ITEM002	Item test integration10	NPCT1-ENG-STORE	1.00	SPRPART		NPCT1	ACTIVE

 - 2.a **Header Bar**, give information that user is in the **Inventory** application.
 - 2.b **Item** field, to filter inventory data by Item code.
 - 2.c **Description** field, to filter inventory data by description.
 - 2.d **Storeroom** field, to filter inventory data by storeroom.
 - 2.e **Commodity Group** field, to filter inventory by commodity group.
 - 2.f **Rotating** field, to filter inventory data by Rotating status of the item.
 - 2.g **Site** field, to filter inventory data by Site
 - 2.h **Status** field, to filter inventory data by Status

3 Choose one inventory data to view the details



Item: ITEM001 >> Lamp - remote warning light aa Site: NPCT1 Attachments

Store room: NPCT1-ENG-1 >> Engineering Store room Status: ACTIVE Issue Unit: UNIT

Default Bin: Consignment? Condition Enabled? Rotating?

Available Balance Summary Other Balance Summary Information Issue History

Current Balance: 409.00 Quantity Currently Reserved: 125.00 Last Issue Date: 3/16/16 6:33 PM

Quantity in Holding Location: 0.00 Hard Reserved Quantity: 125.00 Year to Date: 18.00

Quantity Available: 284.00 Soft Reserved Quantity: 0.00 Last Year: 0.00

2 Years Ago: 0.00

Inventory Costs Filter > 1 - 1 of 1

Condition Code	Description	Condition Rate	Average Cost	W Average Cost	W Average Cost 2	W Initial Balance	Last Receipt Cost
100		200.00	500.00	0.04	4.00	1,000.00	

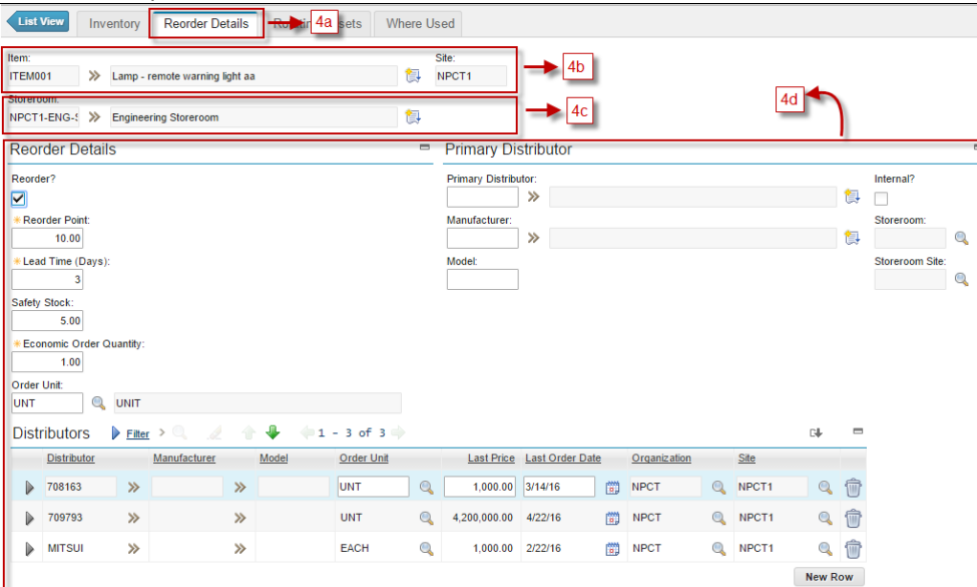
Inventory Balances Filter > 1 - 1 of 1

Bin	Condition Code	Current Balance	Physical Count	Physical Count Date	Reconciled?
		409.00	400.00	3/11/16 10:09 AM	✓

3.a **Inventory** tab, to view the details of the inventory data

3.b Detail information of the inventory data, such as current balance, quantity available, issue unit, inventory cost, and etc.

4



Item: ITEM001 >> Lamp - remote warning light aa Site: NPCT1

Store room: NPCT1-ENG-1 >> Engineering Store room

Reorder? ☒ Reorder Point: 10.00 Lead Time (Days): 3 Safety Stock: 5.00 Economic Order Quantity: 1.00 Order Unit: UNIT

Primary Distributor: Internal? ☐ Store room: Store room Site: Store room Site

Distributors Filter > 1 - 3 of 3

Distributor	Manufacturer	Model	Order Unit	Last Price	Last Order Date	Organization	Site
708163			UNT	1,000.00	3/14/16	NPCT	NPCT1
709793			UNT	4,200,000.00	4/22/16	NPCT	NPCT1
MITSUMI			EACH	1,000.00	2/22/16	NPCT	NPCT1

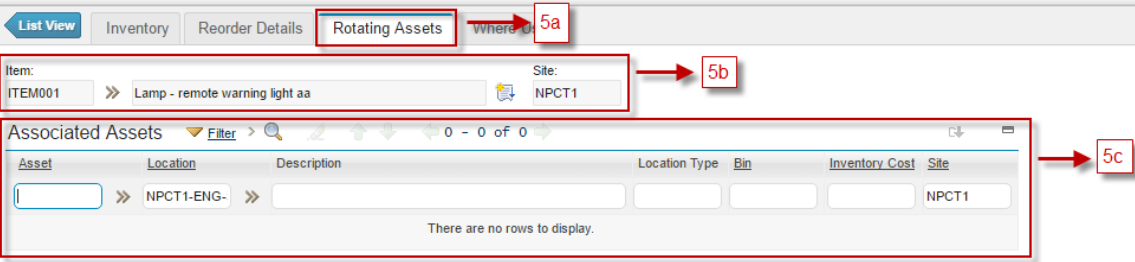
4.a **Reorder Details** tab, to view and update reorder configurations

4.b Item information of the inventory data

4.c Store room information of the inventory data

4.d Detail information of reorder configuration toward the inventory data, such as reorder point, lead time, safety stock, economic order quantity, and etc.

5

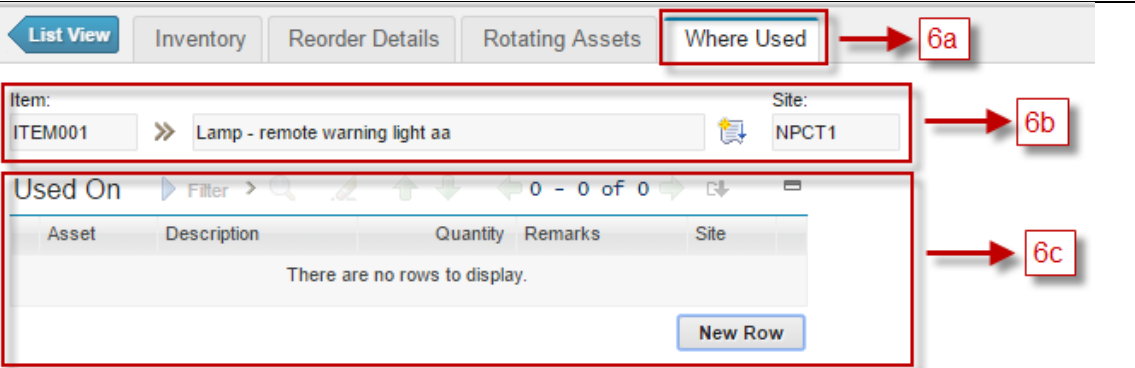


5.a **Rotating Assets** tab, to view all rotating asset of the item (this tab is useful for rotating item only)

5.b Item information of the inventory data

5.c Listed all rotating assets related with item based on filtering result

6



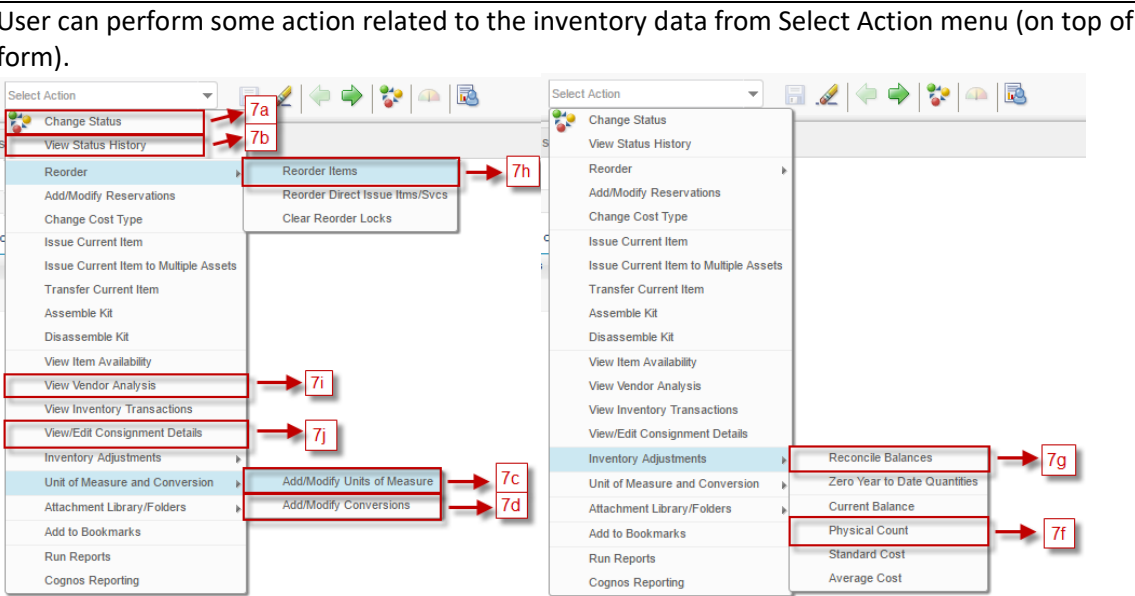
6.a **Where Used** tab, to view all asset where the item is registered as their spare part.

6.b Item information of the inventory data

6.c Listed all asset where the item is registered as their spare part.

7

User can perform some action related to the inventory data from Select Action menu (on top of the form).



7.a **Change Status** : action for change status of inventory data

7.b **View Status History**, to view the change status history of inventory data.

7.c	Add/Modify Units of Measure , to view and modify units of measure master data
7.d	Add/Modify Conversions , to view and modify conversions master data
7.e	Add/Modify Reservation , to view and modify reservation toward all inventory data related to work.
7.f	Physical Count , to input physical count result related to Stock Opname process.
7.g	Reconcile Balances , to change the current balance of specific inventory data based on physical count result.
7.h	Reorder Items , to perform reorder item manually, resulting in PR generation.
7.i	View Vendor Analysis , to view vendor performances related to inventory data.
7.j	View/Edit Consignment Details , to marks the inventory data as consignment

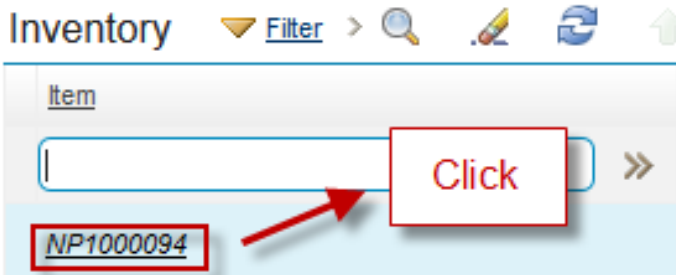
2.2.1 Reorder Item

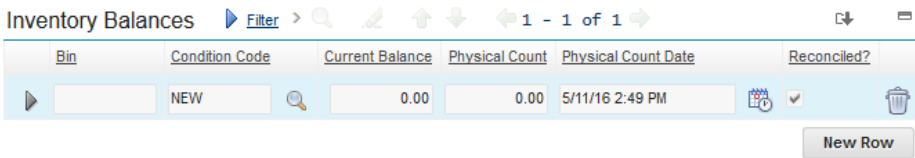
With the daily use of inventory items, COMP1 must replenish inventory stock to stay functional by replacing used inventory stock by reserving and reordering items that COMP1 need. By configuring reorder details towards specific item in a storeroom, stock availability of those items can be maintained effectively and efficient. **Safety Stock**, **Reorder Point**, **Lead Time**, and **Economic Order Quantity** can be defined for each item in storeroom to support this reorder calculations. As an addition, EMS can give suggestion to user in defining **Reorder Point** and **Economic Order Quantity** through:

- Inventory ROP (Reorder Point) Analysis report.
- Inventory EOQ (Economic Order Quantity) Analysis report.

As result of reorder configuration on inventory data, when a reorder needs to be performed, PR can be generated for related items with suggested value. This PR generation can be initiated *manually* via **Reorder Item** action, or *automatically* by system.

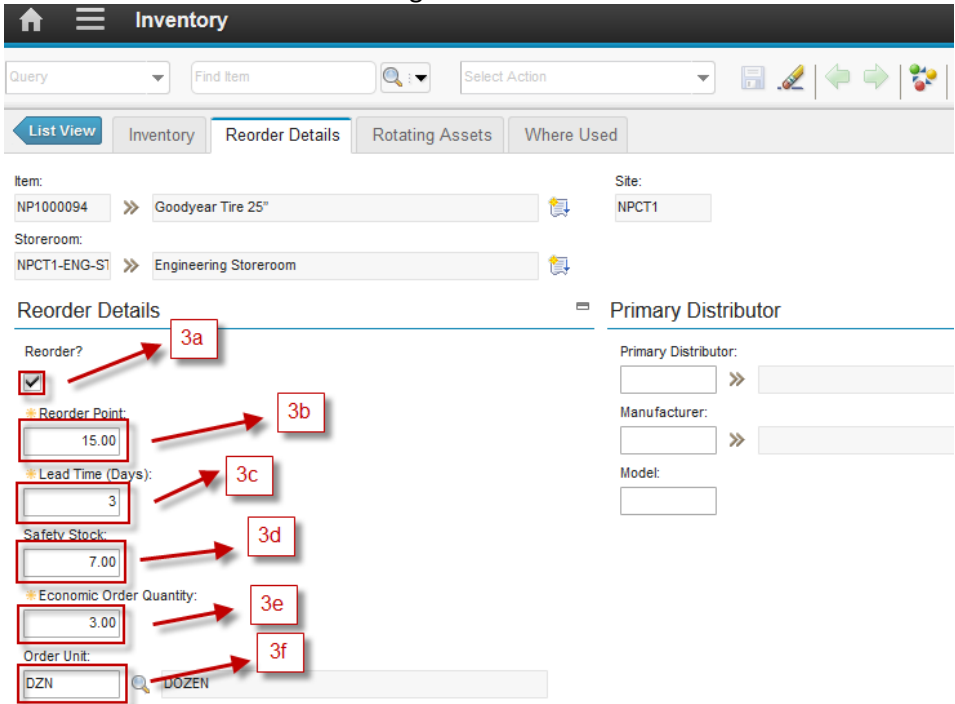
Below are details information about reorder configuration and manually reorder process. Reorder configuration can be performed by *Engineering Manager*, while **Reorder Item** action can be performed by *Store Man*.

1	As <i>Engineering Manager</i> , from Inventory application, choose one of the inventory data available. 
---	---

- 2 To enable reorder process, make sure that **Inventory Balances** table in **Inventory** tab has at least 1 line. If not, click **New Row** button to add a line, then save .
 

Inventory Balances  >     1 - 1 of 1  

Bin	Condition Code	Current Balance	Physical Count	Physical Count Date	Reconciled?
	NEW 	0.00	0.00	5/11/16 2:49 PM	 


- 3 Go to **Reorder Details** tab to configure the reorder.
 

Item: NP1000094 >> Goodyear Tire 25" Site: NPCT1
 Storeroom: NPCT1-ENG-ST >> Engineering Storeroom

Reorder Details

Reorder? ☒ **3a**

Reorder Point: 15.00 **3b**

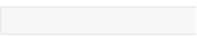
Lead Time (Days): 3 **3c**

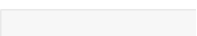
Safety Stock: 7.00 **3d**

Economic Order Quantity: 3.00 **3e**

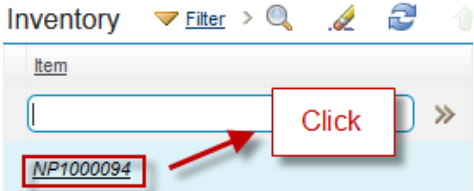
Order Unit: DZN  DOZEN **3f**

Primary Distributor

Primary Distributor:  >> 

Manufacturer:  >> 

Model: 

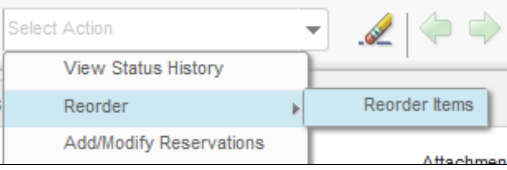
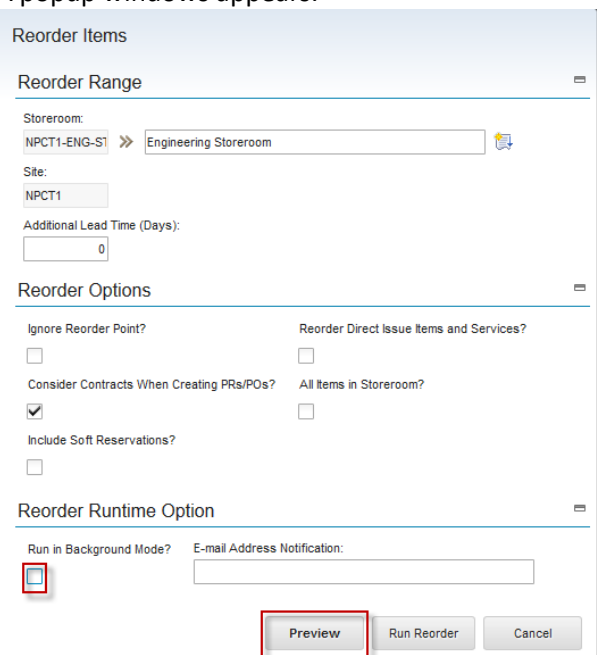
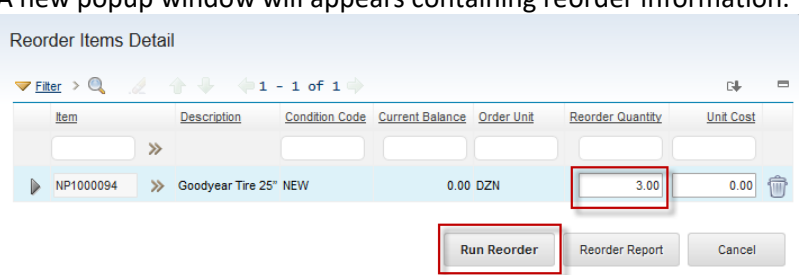
3.a **Reorder** checkbox, indicates that reorder process can be performed to this inventory data.
 3.b **Reorder Point** (ROP) field
 3.c **Lead Time (Days)**, fill with estimated time from ordering item to receiving the item.
 3.d **Safety Stock** field, fill with safety stock value. Reorder point should be higher than safety stock, because the best practice is reorder should be performed before the stock touch the safety stock value.
 3.e **Economic Order Quantity** (EOQ) field
 3.f **Order Unit** field, the unit of measure of EOQ
- 4 After all reorder configuration is done, now manual reorder process can be performed toward those inventory data by *Store Man*. Choose the inventory data where reorder process want to be performed.
 

Inventory  >    

Item

  >>

NP1000094 

	Take notes on the Available Balance Summary section. <u>Available Balance Summary</u>  Current Balance: 0.00 Quantity in Holding Location: 0.00 Quantity Available: 0.00
5	From Select Action menu, choose Reorder - Reorder Item. 
6	A popup windows appears.  Uncheck the Run in Background Mode checkbox, so the reorder process can be showed. Then click Preview button.
7	A new popup window will appears containing reorder information.  Notify the Reorder Quantity suggested by the system, modify the value as necessary. Click Run Reorder button to perform the reorder process.

8	The PR created from this reorder process is showed. Reorder Results Total of Items Reordered: 1 PRs Generated From: PR/ENG/10122 PRs Generated To: PR/ENG/10122 POs Generated From: POs Generated To: OK
---	--

2.2.2 Stock Opname

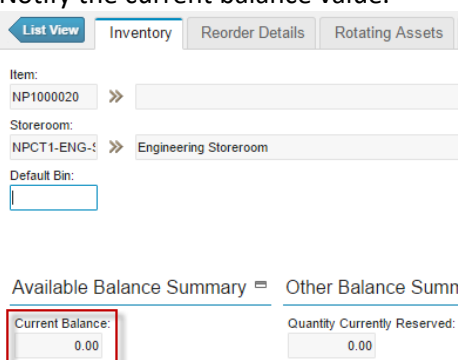
Stock Opname is reconciliation process between stocks in the system against physical stocks in storeroom. The proses is performed by counting the real stock in storeroom, then apply it to the system, with the variance value will be recorded in Shrinkage Account of the storeroom.

When there is a variance between real stock and system stock, user can register the real stock via **Physical Count** action available in **Inventory** application. This action can be accessed by *Store Man*. Once the physical stock has been inputted, it can be applied as new system current balance via **Reconcile Balance** action, which can be performed by *Accounting*. Stock Opname can only be executed toward **non-rotating** items.

Below are details steps about stock opname process via **Inventory** application.

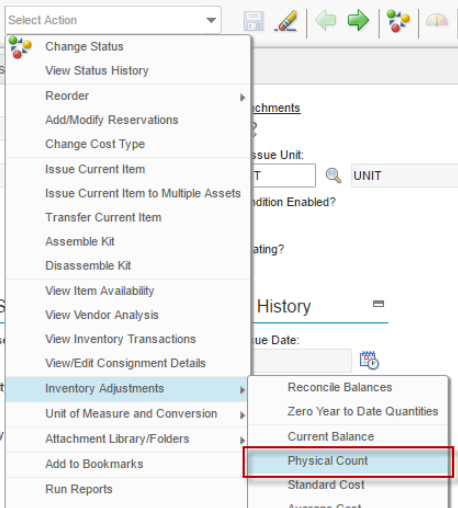
1	First is Physical Count report by <i>Store Man</i>. From Inventory application, choose one of the inventory data available. <u>Item</u> NP1000020 >> NP1000020 ➔ Click
---	---

3 Notify the current balance value.



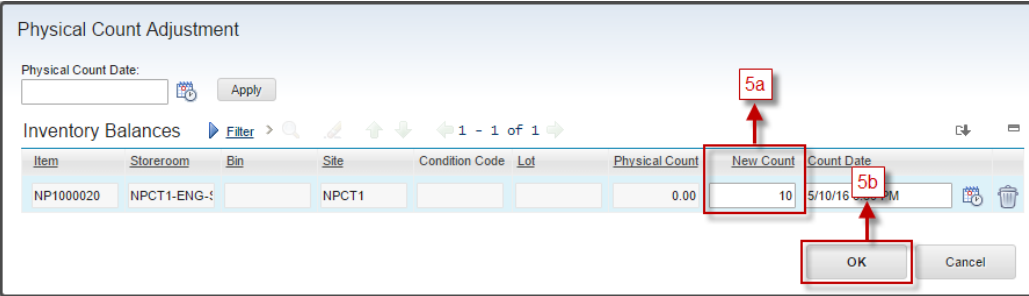
The screenshot shows the 'Inventory' tab with fields for Item (NP1000020), Storeroom (NPCT1-ENG-1), and Default Bin. Below these, the 'Available Balance Summary' section shows 'Current Balance' as 0.00, which is highlighted with a red box. The 'Quantity Currently Reserved' is also 0.00.

4 Choose **Select Action – Inventory Adjustments – Physical Count**



The screenshot shows the 'Select Action' dropdown menu. The 'Inventory Adjustments' option is selected, and the 'Physical Count' sub-option is highlighted with a red box. Other options include Change Status, View Status History, Reorder, Add/Modify Reservations, Change Cost Type, Issue Current Item, Issue Current Item to Multiple Assets, Transfer Current Item, Assemble Kit, Disassemble Kit, View Item Availability, View Vendor Analysis, View Inventory Transactions, View/Edit Consignment Details, Reconcile Balances, Zero Year to Date Quantities, Current Balance, Standard Cost, and Average Cost.

5 Show windows “Physical Count Adjustment”.

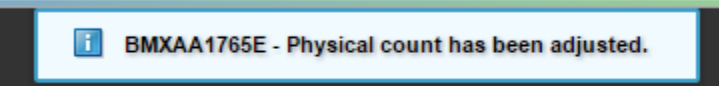


The screenshot shows the 'Physical Count Adjustment' window. It includes a 'Physical Count Date' field, an 'Apply' button, and a table of 'Inventory Balances'. The table has columns for Item, Storeroom, Bin, Site, Condition Code, Lot, Physical Count, New Count, and Count Date. The 'New Count' field for item NP1000020 is highlighted with a red box and labeled '5a'. The 'OK' button is highlighted with a red box and labeled '5b'.

5.a Filled field New Count with number of physical count

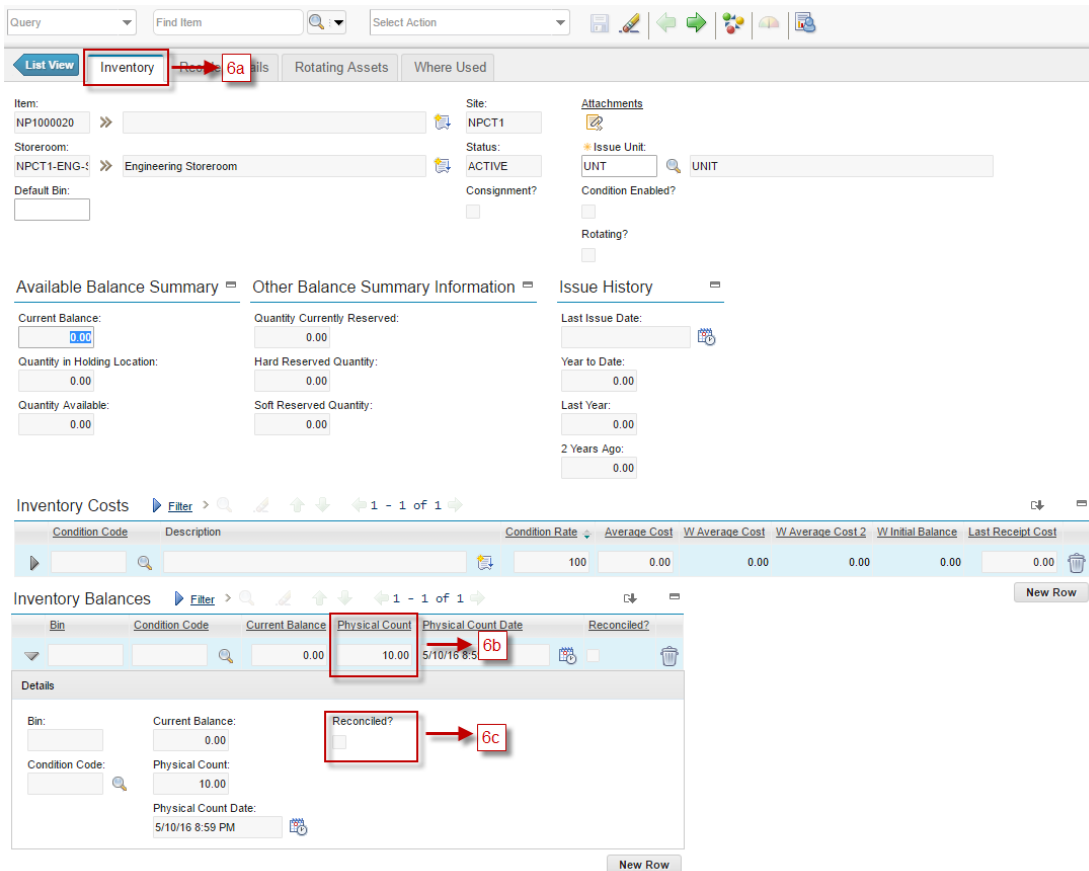
5.b Click “OK” to the next progress

In the top page show information about “Physical count has been adjusted”



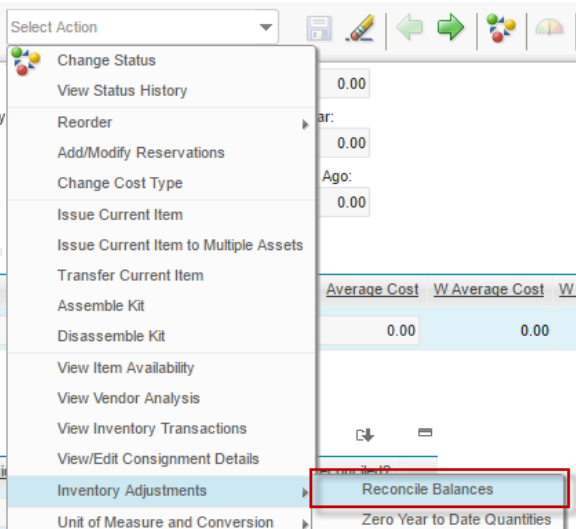
The screenshot shows a system message bar with the text 'BMXAA1765E - Physical count has been adjusted.'

- 6 The result of this process showed in **Inventory Balance** table in **Inventory** tab (6a). The **Physical Count** (6b) is updated, and the **Reconciled** (6c) checkbox is unchecked, means that the physical count is not reconciled to the current balance yet.

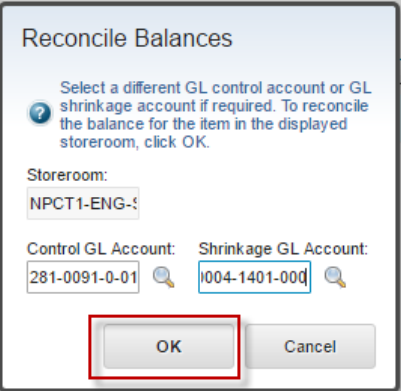
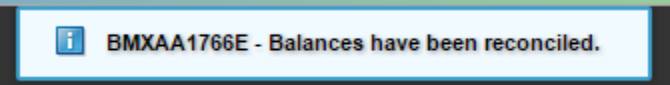
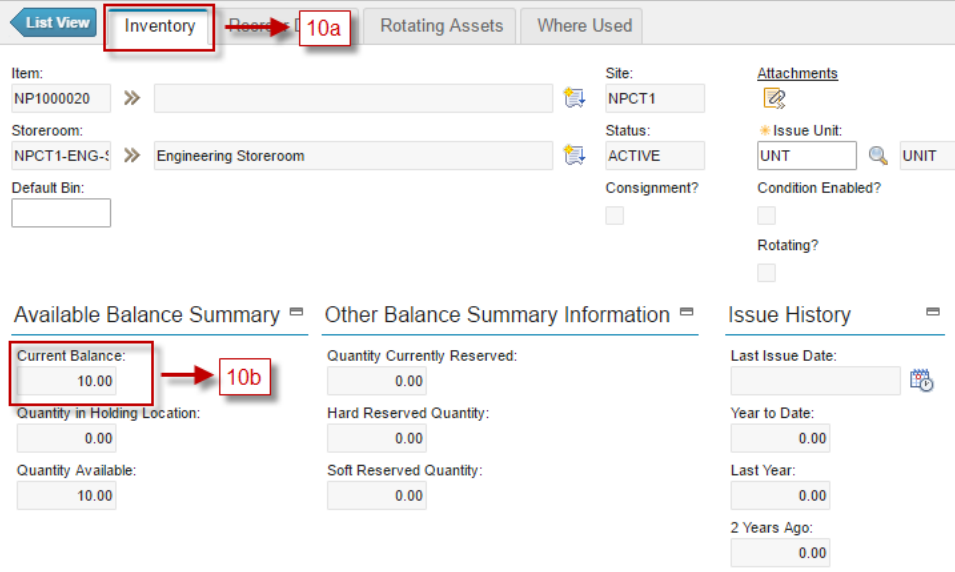


The screenshot displays the Inventory Management interface. At the top, there's a navigation bar with tabs: List View, Inventory (6a), Reconcile, Rotating Assets, and Where Used. The Inventory tab is active, showing details for Item: NP1000020, Site: NPCT1, Storeroom: NPCT1-ENG-1, and Status: ACTIVE. Below this, there's a section for Available Balance Summary and Other Balance Summary Information. The Available Balance Summary shows Current Balance: 0.00, Quantity in Holding Location: 0.00, and Quantity Available: 0.00. The Other Balance Summary Information shows Quantity Currently Reserved: 0.00, Hard Reserved Quantity: 0.00, and Soft Reserved Quantity: 0.00. Below these, there's a table for Inventory Costs and a table for Inventory Balances. The Inventory Balances table has columns: Bin, Condition Code, Current Balance, Physical Count (6b), Physical Count Date, and Reconciled?. The Physical Count (6b) is highlighted with a red box. Below the table, there's a Details section for the selected row, showing Bin, Current Balance, Condition Code, Physical Count, Physical Count Date, and Reconciled? (6c). The Reconciled? checkbox is highlighted with a red box.

- 7 Next process is to reconcile the balance by *Accounting*. As *Accounting*, in the inventory data, choose action **Inventory Adjustments – Reconcile Balances**



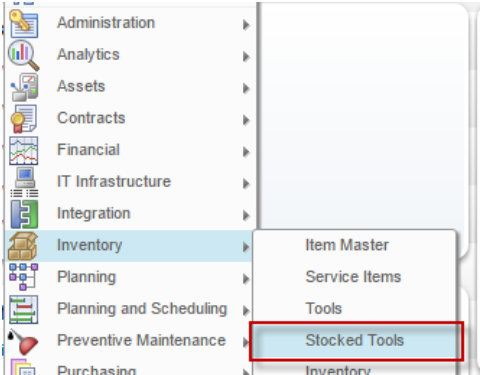
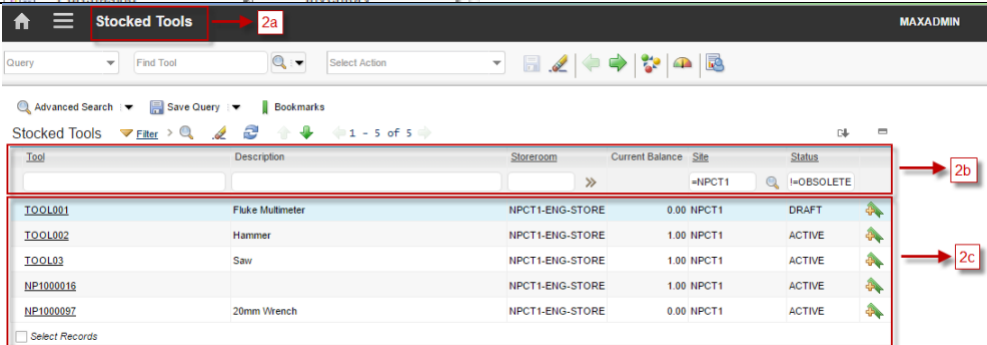
The screenshot shows the 'Select Action' dropdown menu open, displaying various actions. The 'Inventory Adjustments' option is selected, and the 'Reconcile Balances' sub-option is highlighted with a red box. The background shows the same Inventory Management interface as in the previous screenshot, with the 'Inventory' tab active and the 'Physical Count' (6b) and 'Reconciled?' (6c) fields visible.

8	Show window “Reconcile Balance”  Then click “OK”
9	In the top page view information “Balances have been reconciled” 
10	In the Inventory Tab (10a), Current Balance (10b) is updated 

2.3 Stocked Tools Application

Stocked tools is a listing of all tools that are stored in storerooms (called stocked tool data). Using the **Stocked Tools** application, user can track tool balances down to the bin and lot level for a storeroom. Reorder process does not apply here, considering tools is not consumed. Stocked tools data created when user perform **Add Tool to Storeroom** action from **Tool** application, or created automatically when a tool is purchased and received into specific storeroom for the first time.

Followings are details about **Stocked Tools** applications accessibility:

- 1 From **Go To** menu, choose **Inventory – Stocked Tools**

- 2
 

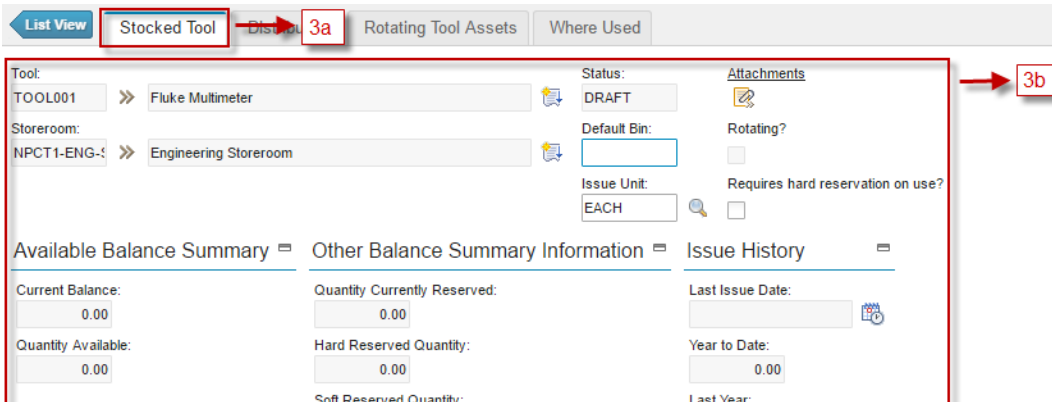
2.a **Header Bar** will give information that user is in the **Stocked Tools** application.

2.b **Filter** stocked tool data by **Tool** code, **Description**, **Storeroom**, **Site**, and **Status**

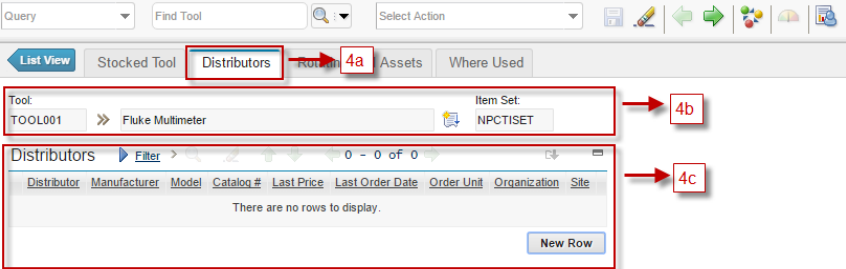
2.c Listed all stocked tools data based on filtering result
- 3 Choose one stocked tool data to view the details
 

3.a **Stocked Tool** Tab, view detail of stocked tool data, such as current balance, quantity available, etc.

3.b Details information about stocked tool data



4

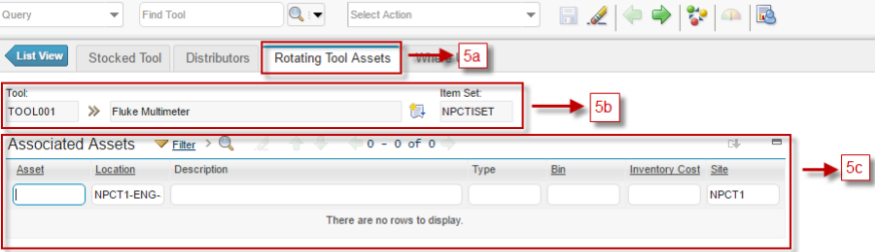


4.a **Distributors** Tab, view all distributors of related tool

4.b Tool information of stocked tool data

4.c Listed distributors related with tool

5



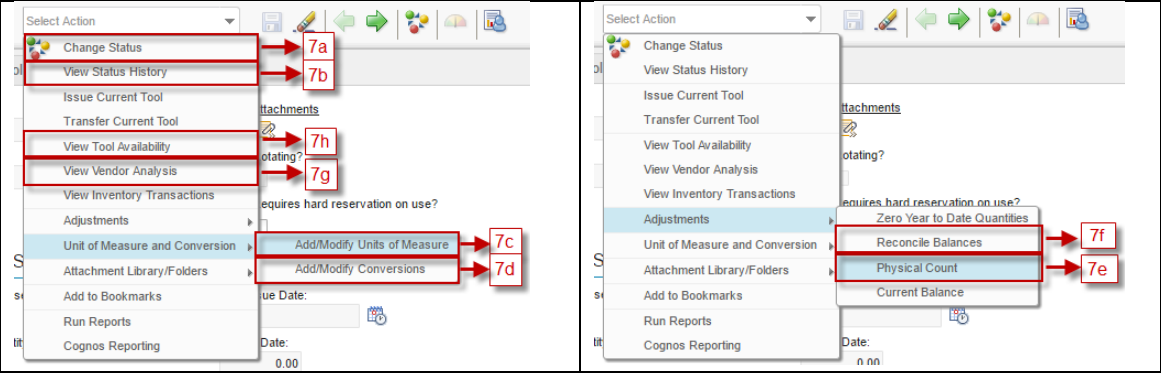
5.d **Rotating Tool Assets** Tab, to view all rotating asset of the tool (this tab is useful for rotating tool only)

5.e Tool information of the stocked tool data

5.a Listed all rotating assets related with tool based on filtering result

6

User can perform some action related to the stocked tools data from Select Action menu (on top of the form).



7.a **Change Status** : action for change status of stocked tool data

7.b **View Status History**, to view the change status history of stocked tool data.

7.c **Add/Modify Units of Measure**, to view and modify units of measure master data

7.d **Add/Modify Conversions**, to view and modify conversions master data

7.e **Physical Count**, to input physical count result related to Stock Opname process.

7.f **Reconcile Balances**, to change the current balance of specific stocked tool data based on physical count result.

7.g **View Vendor Analysis**, to view vendor performances related to stocked tool data.

7.h **View Tool Availability**, to view availability of the tool.

2.3.1 Stock Opname

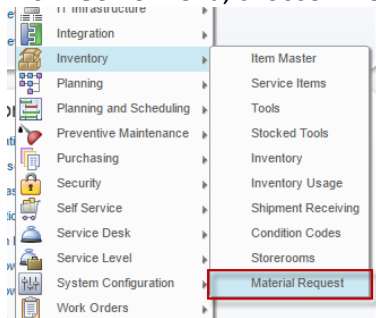
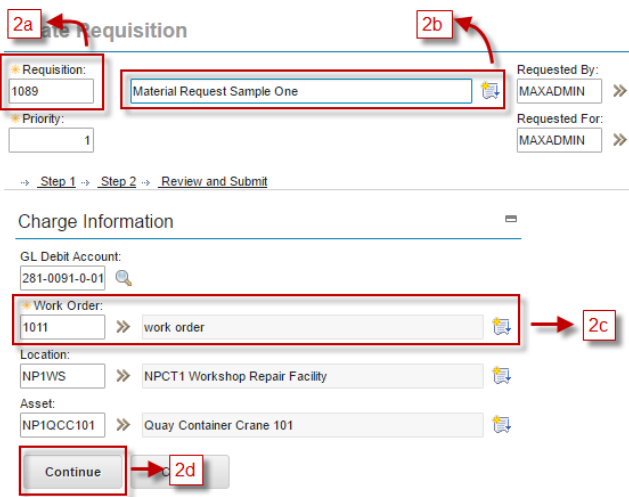
Stock Opname can be performed toward tools, via **Stocked Tools** application. The process is the same as Stock Opname in **Inventory** application. **Physical Count** action can be accessed by *Store Man*, while **Reconcile Balance** by *Accounting*. Stock Opname can only be executed toward **non-rotating** tools.

2.4 Material Request Application

Material Request application is another way available to request item or services. Material Request is needed to request additional material or service regarding maintenance work, usually unplanned materials or services. Material Request can be created by *Service Engineer*. Approval by *CHE Engineer* is needed for further response toward a material request.

2.4.1 Material Request Submission

Below are steps to create and submit a material request.

1	From Go To menu, choose Inventory – Material request. 
2	Material Request form is displayed.  2.a The <i>Requisition</i> column is filled with a new auto-generated code. 2.b Filled field about description of material request 2.c Choose one Work Order form listed work order and Information about GL Debit Account, Location, Asset automatically filled if the Work Order have this information. 2.d Click “Continue” button for the next step

3 Click **New Row** button.

Create Requisition

* Requisition: 1090 Material Request Sample One
 * Priority: 1
 Requested By: MAXADMIN
 Requested For: MAXADMIN
 → Step 1 → Step 2 → Review and Submit

Requisition Line Items

Line	Quantity	Required Date	Item	Description	Vendor	Store Location
There are no rows to display.						

New Row

Back Continue Cancel

4 Now we will add a line with **Item** type item.

Details

* Line: 1
 * Line Type: Item
 Item: ITEM001 Lamp - remote warning light aa
 Condition Code:
 Classification:
 Class Description:
 * Quantity: 1q
 Issue Unit: UNT
 Unit Cost: 200.00
 * Line Cost: 0.00
 * Currency: IDR

Manufacturer:
 Model:
 Vendor:
 Catalog #:
 Store Location: NPCT1-ENG-
 GL Debit Account: 281-0091-0-01
 * Reservation Type: AUTOMATIC
 * Required Date: 5/11/16 10:24 AM
 Inspection Required?
 Is Distributed?
 Attachments

New Row

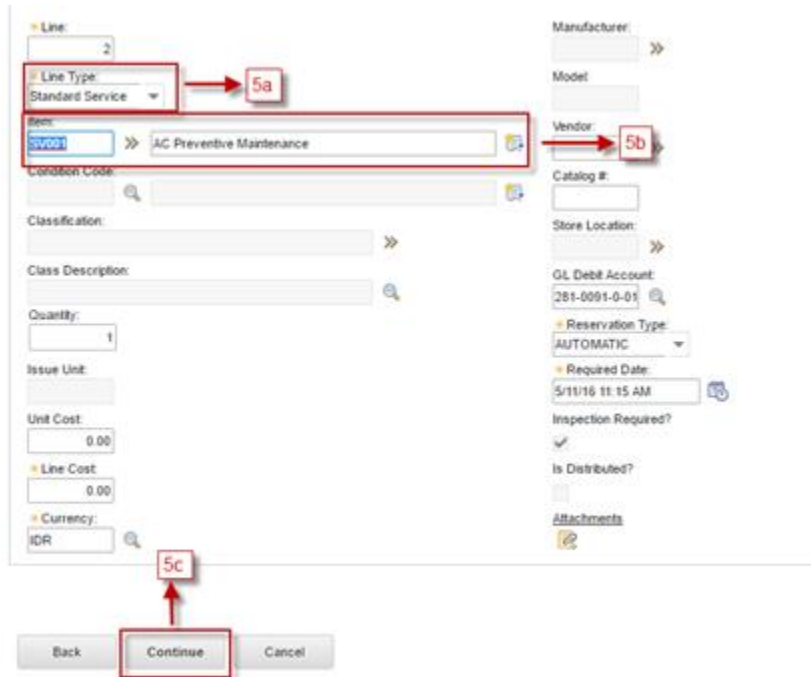
4.a Define the type Items of the item in *Line Type* column.

4.b Item, choose one listed item

4.c Quantity mandatory filled data

4.d **New Row** to add another Requisition Line Items

5 Now we will add a line with **Service** type item.

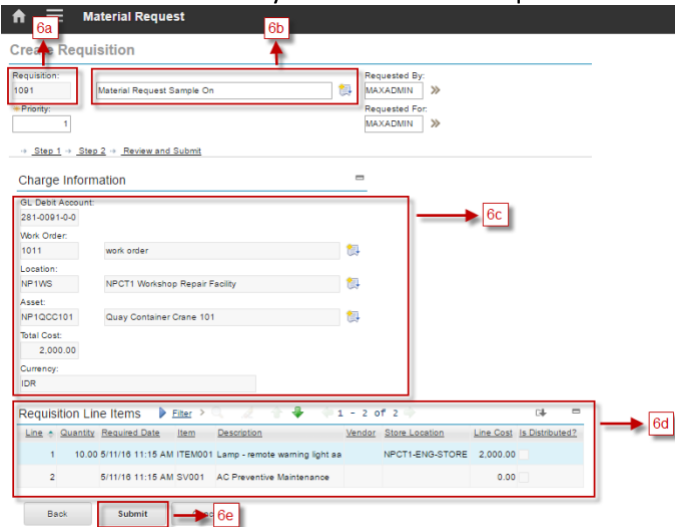


5.a Define the type **Standard Service** of the item in *Line Type* column.

5.b Item, choose one listed Standar Service

5.c Don't forget to provide the quantity. Then Click "Continue" to next step

6 This will show summary of the Material Request that has been inputted.



6.a Information number of Requisition

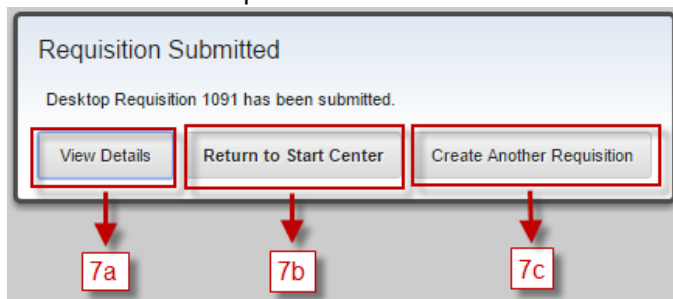
6.b Information description of Requisition

6.c Summary Information from step 2.

6.d Summary Information from step 3, 4, and 5.

6.e Click "Submit" button to submit the Material Request.

7 Show windows “Requisition Submitted”



7.a **View Details**, for view details of the requisition

7.b **Return to Start Center**, to return to the Home menu

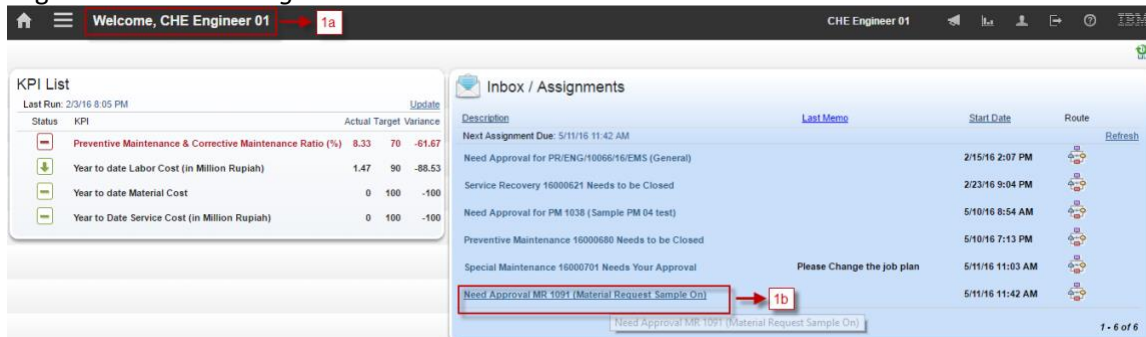
7.c **Create Another Requisition**, for create another requisition

2.4.2 Material Request Approval by CHE Engineer.

Below are steps to approve a material request. For material request rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of Job Plan in **Work Management (part 1)** training material.

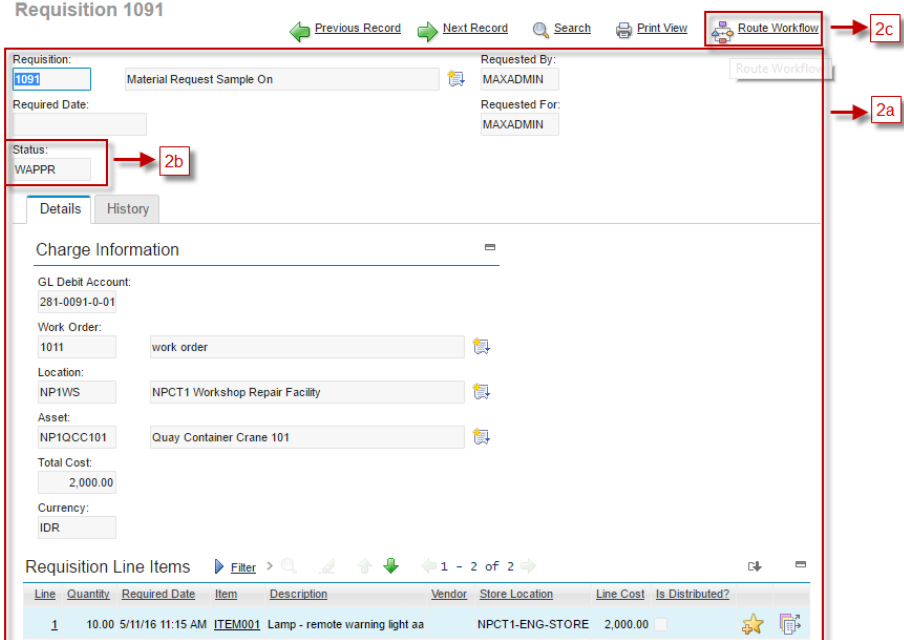
As a note, request revision for PM Work Schedule will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for approval.

1 Login user with *CHE Engineer*

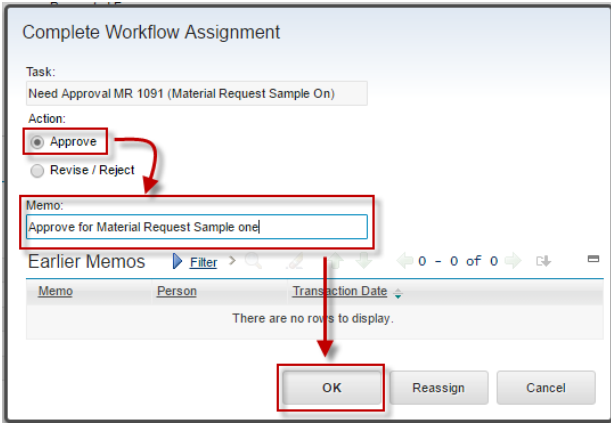


1.a Information about user login

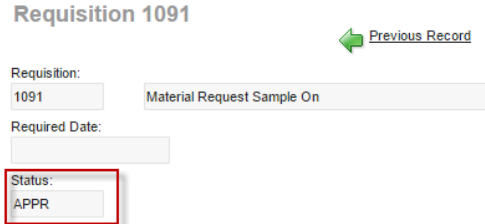
1.b In Inbox/Assignment portlet, an assignment to approve material request is available. Click the assignment to go to the Material Request data.

- 2
 - 2.a Details material request
 - 2.b Information status material request (WAPPR)
 - 2.c Route workflow, to approve this material
- 3

Windows show “Complete Workflow Assignment”. Choose Approve and filled field Memo and the Click “OK”


- 4

Status change APPR (Approve)



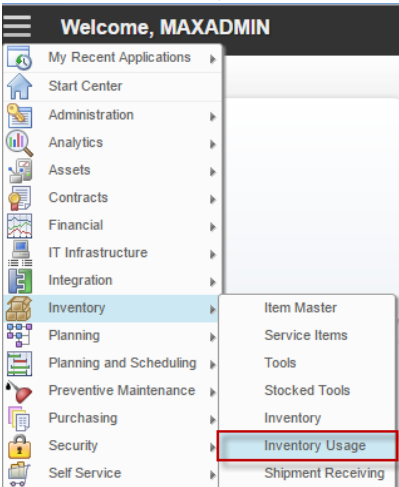
2.5 Inventory Usage Application

Usage document to record inventory usage and return are maintained in **Inventory Usage** application. The status of inventory usage document is as follows:

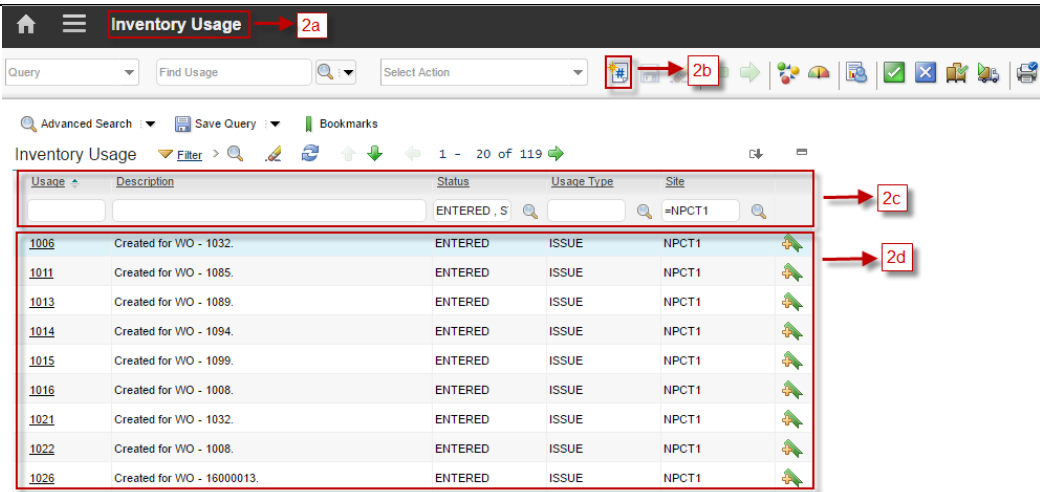
- **ENTERED** : The default status for a usage document record when it is created.
- **COMPLETE** : Indicates that the issue or return of items is complete. When user change the status to **COMPLETE**, the transaction is executed and the inventory balance is adjusted.
- **CANCELLED** : The issue or return is cancelled.

Followings are details about **Inventory Usage** applications accessibility:

1
From **Go To** menu, choose **Inventory – Inventory Usage**



2



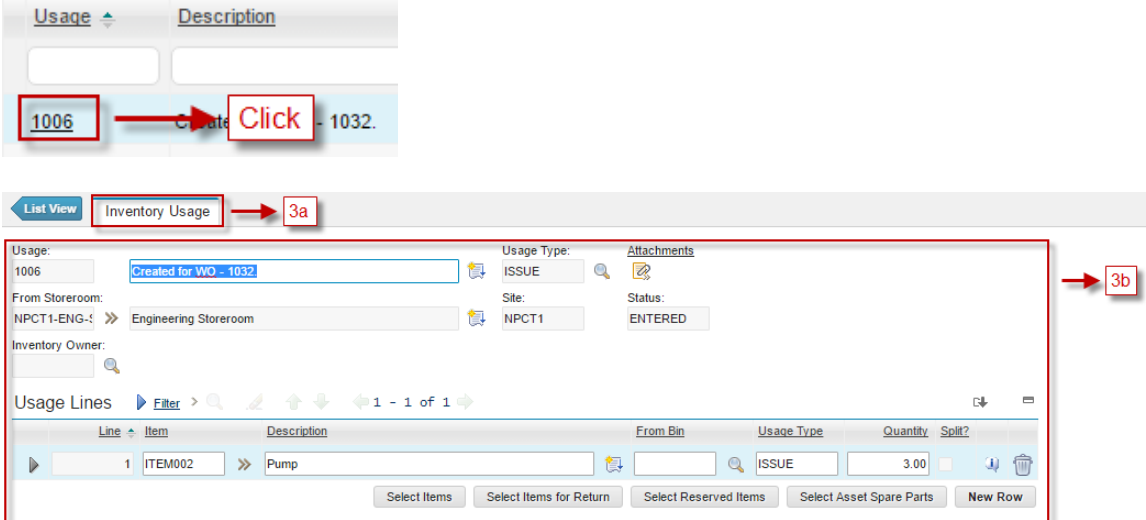
2.a **Header Bar** will give information that user is in the **Inventory Usage** application

2.d **New Usage** button, to add new inventory usage

2.e **Filter** usage document by **Usage Number**, **Description**, **Status**, **Usage Type** and **Site**

2.f List of all usage document based on filtering result.

3 Choose one inventory usage document to view details



Usage: 1006 Description: Created for WO - 1032

Usage Type: ISSUE Site: NPCT1 Status: ENTERED

From Storeroom: NPCT1-ENG-1 Engineering Storeroom

Inventory Owner:

Usage Lines

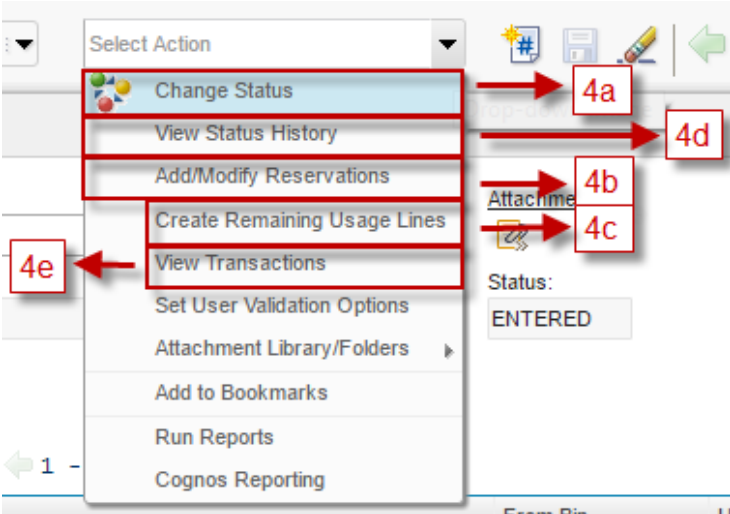
Line	Item	Description	From Bin	Usage Type	Quantity	Split?
1	ITEM002	Pump		ISSUE	3.00	

Select Items Select Items for Return Select Reserved Items Select Asset Spare Parts New Row

3.a **Inventory Usage** Tab, to view and update details information inventory usage

3.b Details information about Inventory Usage

4 User can perform some action related to the inventory usage data from Select Action menu (on top of the form).



Select Action

- Change Status (4a)
- View Status History (4d)
- Add/Modify Reservations (4b)
- Create Remaining Usage Lines (4c)
- View Transactions (4e)
- Set User Validation Options
- Attachment Library/Folders
- Add to Bookmarks
- Run Reports
- Cognos Reporting

Status: ENTERED

4.a **Change Status**, to change status of usage document data

4.b **Add/Modify Reservation**, to view and modify reservation toward all inventory data related to work.

4.c **Create Remaining Usage Lines**, to create another usage document for remaining usage lines that has not been issued for partial issue cases.

4.d **View Status History**, to view usage document status history.

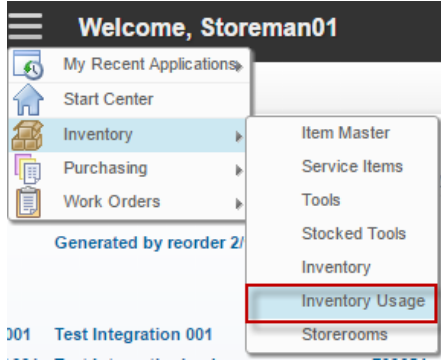
4.e **View Transaction**, to view inventory transaction history toward this usage document.

2.5.1 Inventory Issue

Inventory usage data is created automatically by system when an item reservation occurs, to issue the item. But, user can manually create new inventory usage data as necessary to issue an item. This case happens when partial inventory issue occurs (Only some of required items are available at the moment and issued. The rest will be issued later after they are available). As additional information, every lines of inventory issue should contain **Work Order** information. Inventory issue can be performed by *Store Man*.

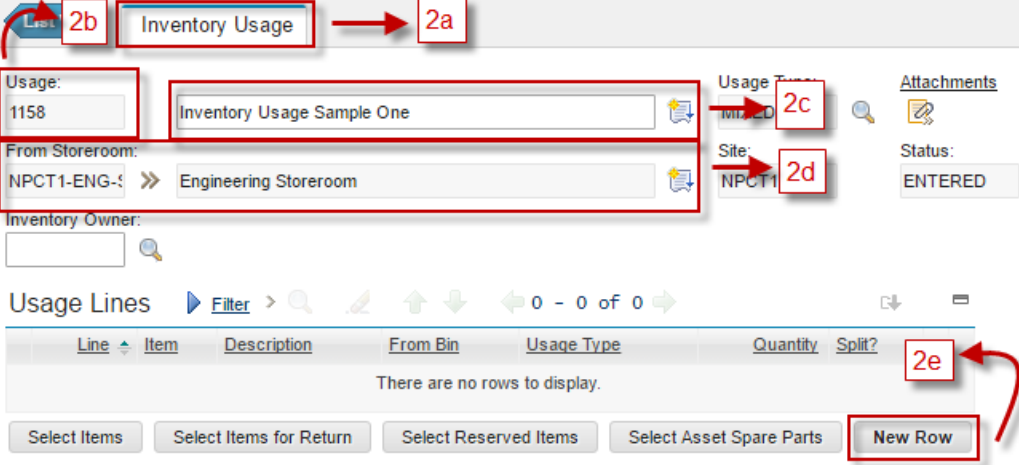
Below are steps to create and execute an inventory issue.

1 Login as Store Man. From **Go To** menu, choose **Inventory – Inventory Usage**



Click New Inventory Usage  to add new inventory usage

2



2.a Inventory Usage Tab, filled information about new inventory usage tab

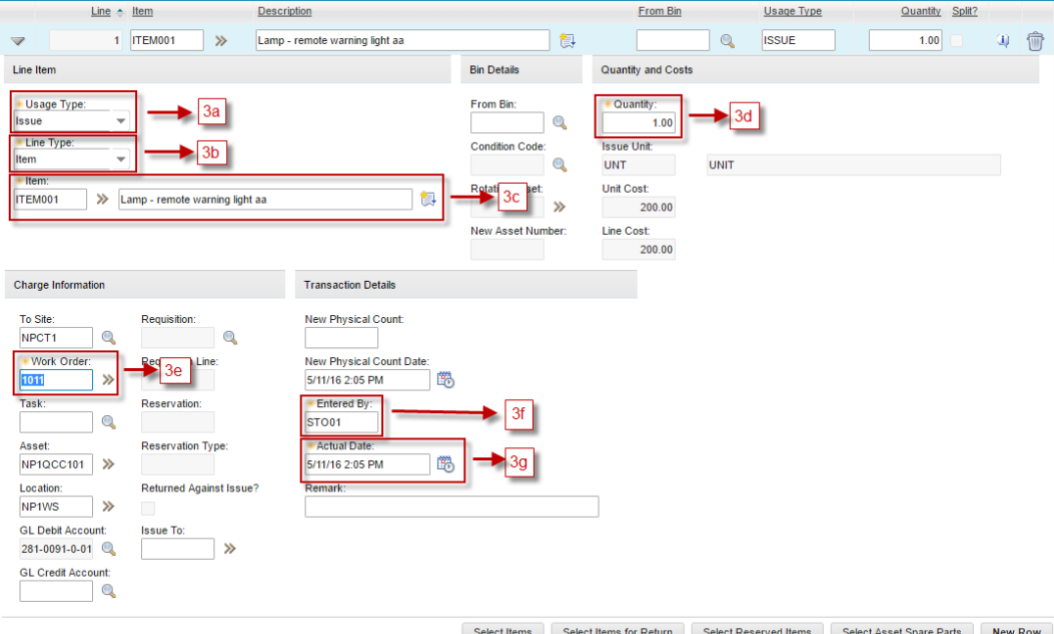
2.b Usage Code, running number usage automatically filled by system

2.c Filled field about details Inventory Usage

2.d Filled field with Storeroom listed

2.e New Row for added usage line

3



3.a Usage type, mandatory requirement the initial value is Issue

3.b Line Type, mandatory requirement the initial values is Item

3.c Item, mandatory requirement. Filled with one listed item.

3.d Quantity, mandatory requirement the initial values is one (1).

3.e Work Order, mandatory requirement. Filled with one listed work order. Field Asset, Location, and GL Debit Account automatically filled if date work order complete.

3.f Entered by , automatically filled with user login

3.g Actual Data, automatically filled with date now.

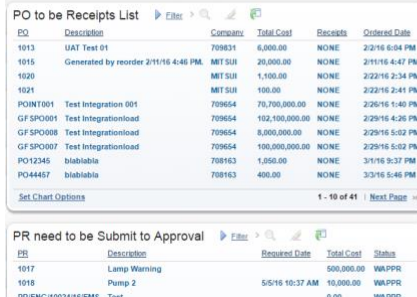
4

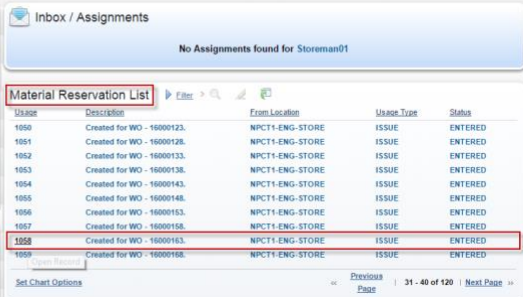
After all field mandatory complete filled. Then save  inventory usage. On the top page show information “Record has been saved”.

 **BMXAA4205I - Record has been saved.**

5

Back to home  menu. Then looked “Material Reservation List”, searching for Inventory usage created before.

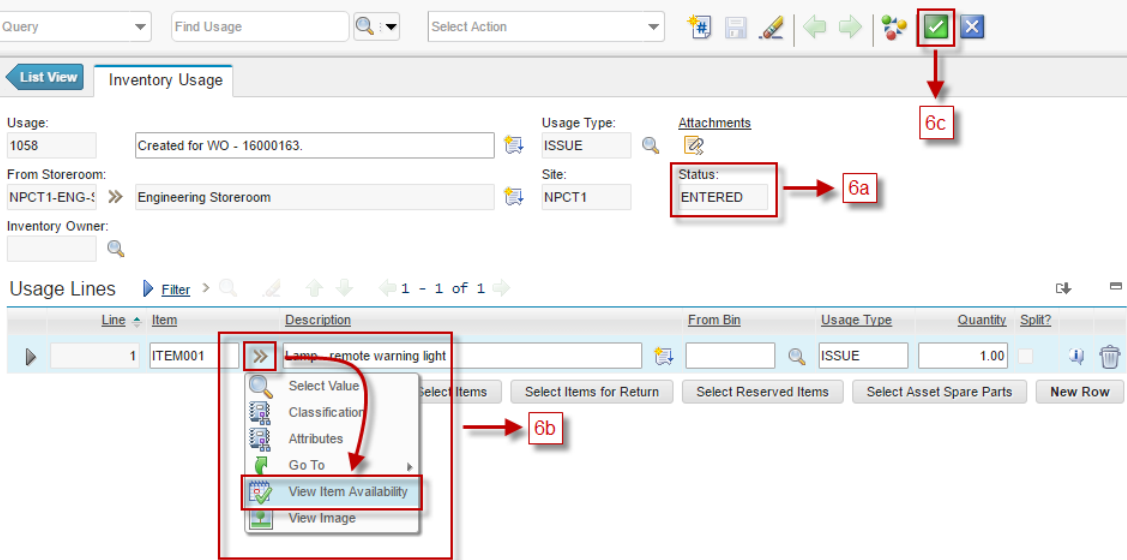




Agile – Proprietary/Confidential

61

6 Show details Inventory Usage.



Query Find Usage Select Action

List View Inventory Usage

Usage: 1058 Created for WO - 16000163. Usage Type: ISSUE Attachments

From Storeroom: NPCT1-ENG-5 Engineering Storeroom Site: NPCT1 Status: ENTERED

Inventory Owner:

Usage Lines Filter 1 - 1 of 1

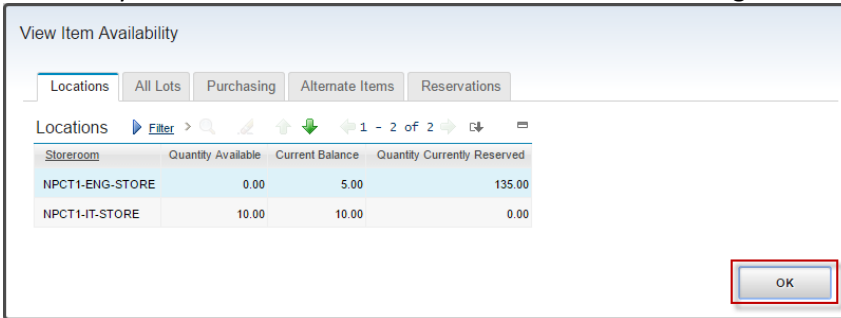
Line	Item	Description	From Bin	Usage Type	Quantity	Split?
1	ITEM001	Lamp - remote warning light		ISSUE	1.00	

Select Value Select Items Select Items for Return Select Reserved Items Select Asset Spare Parts New Row

View Item Availability

6.a Information Status : ENTERED

6.b Show dialog box appears containing related item's stock information. Review the item's availability from this menu. Then click "OK" to close this dialog box.



View Item Availability

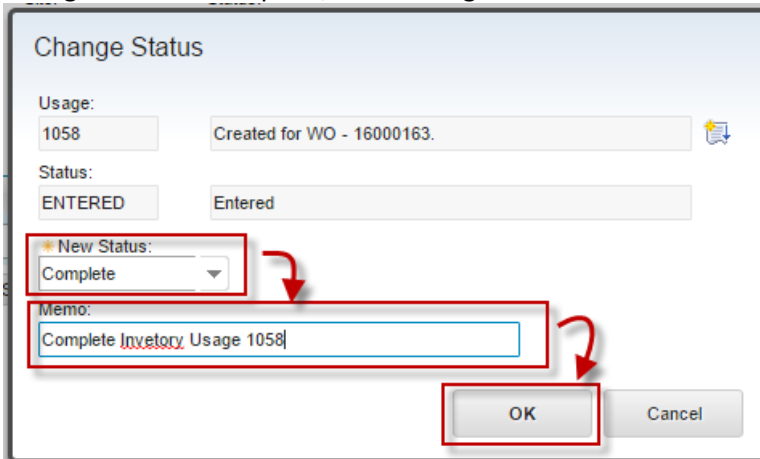
Locations All Lots Purchasing Alternate Items Reservations

Locations Filter 1 - 2 of 2

Storeroom	Quantity Available	Current Balance	Quantity Currently Reserved
NPCT1-ENG-STORE	0.00	5.00	135.00
NPCT1-IT-STORE	10.00	10.00	0.00

OK

6.c Change status to complete, show dialog box



Change Status

Usage: 1058 Created for WO - 16000163.

Status: ENTERED Entered

New Status: Complete

Memo: Complete Inventory Usage 1058

OK Cancel

In the field New Status filled with status Complete, then filled memo. Finally, user can click "OK" button.

7 Status change to COMPLETE



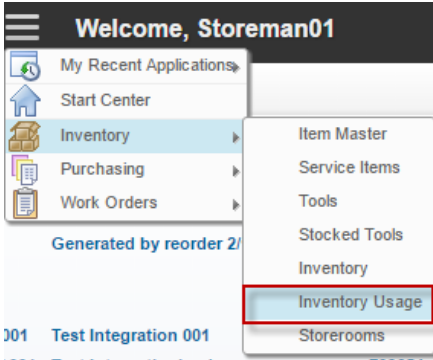
The screenshot shows the 'Inventory Usage' form. The 'Status' field is highlighted with a red box and contains the value 'COMPLETE'. Other fields include 'Usage: 1058', 'Created for WO - 16000163.', 'Usage Type: ISSUE', 'From Storeroom: NPCT1-ENG-1', 'Engineering Storeroom', 'Site: NPCT1', and 'Attachments'.

2.5.2 Inventory Return

To perform inventory return, user should create new inventory usage data manually. Every lines of inventory return should contain **Work Order** information too. Therefore, it is required to perform inventory return first, before closing the work order, so the cost of work order can be adjusted. Inventory return can be performed by *Store Man*.

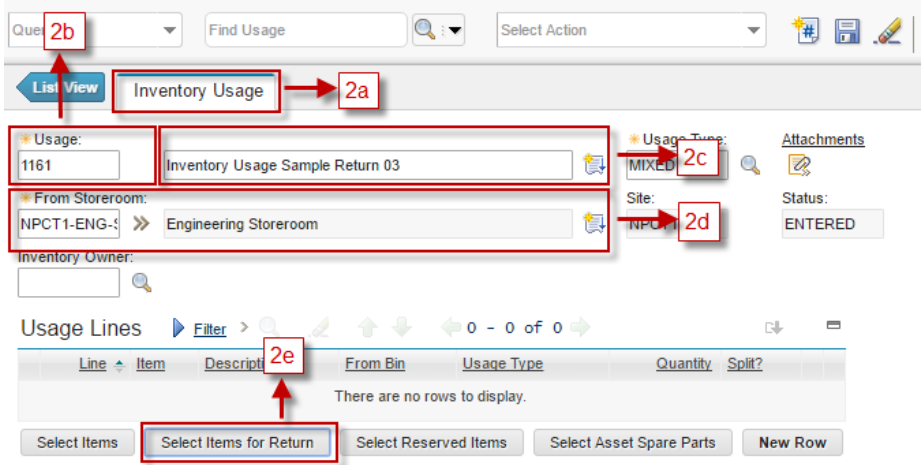
Below are steps to create and execute an inventory return.

1 From Go To menu, choose **Inventory – Inventory Usage**



The screenshot shows the 'Go To' menu with 'Inventory Usage' highlighted in a red box. The menu path is: Inventory > Inventory Usage.

2

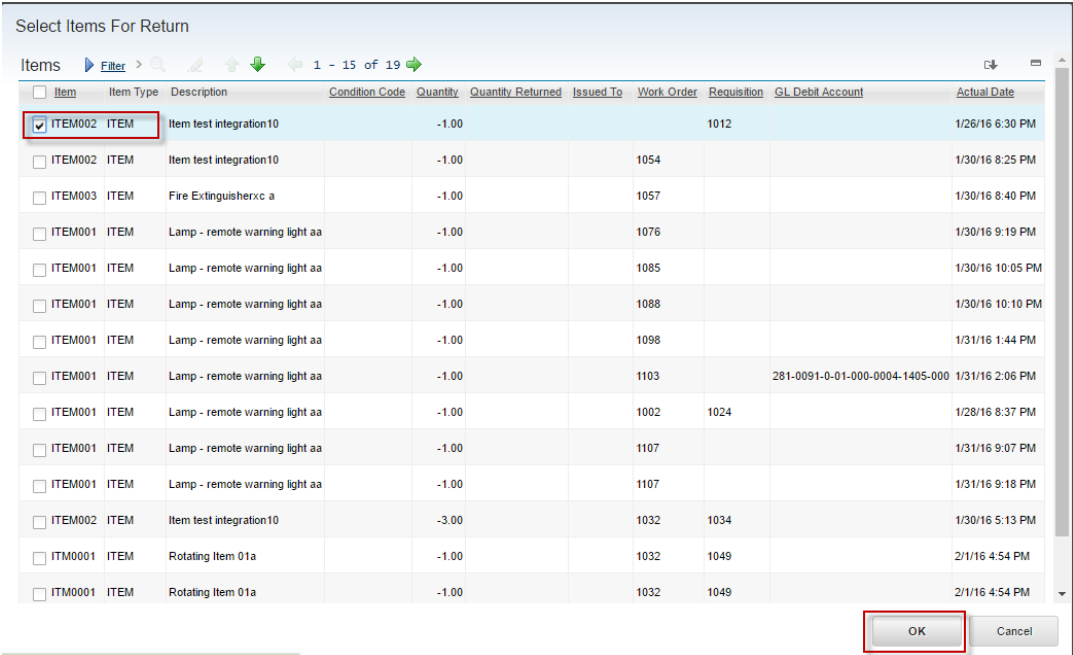
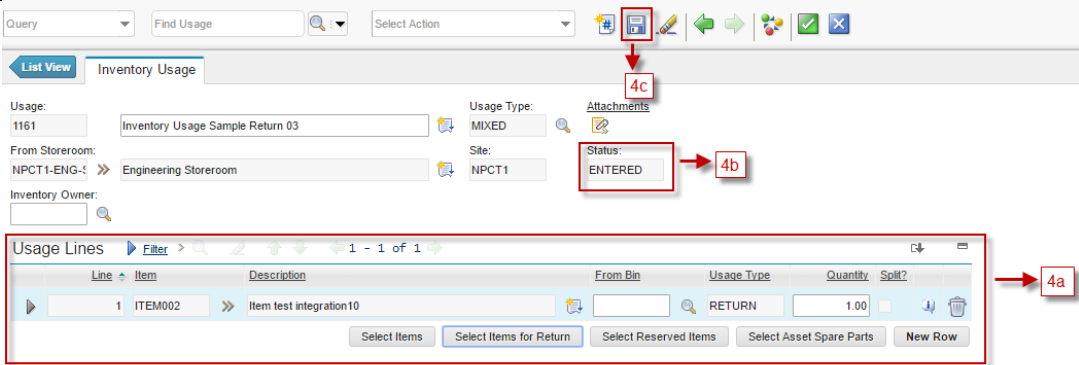
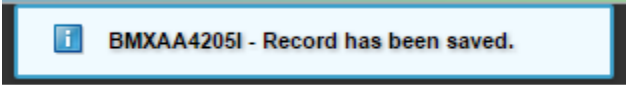


The screenshot shows the 'Inventory Usage' form with several annotations:

- 2b: Points to the 'Find Usage' search bar.
- 2a: Points to the 'Inventory Usage' tab.
- 2c: Points to the 'Usage Type' field, which is set to 'MIXED'.
- 2d: Points to the 'Site' field, which is set to 'NPCT1'.
- 2e: Points to the 'Select Items for Return' button in the 'Usage Lines' section.

 The 'Usage' field is set to '1161' and the 'From Storeroom' is 'NPCT1-ENG-1'. The 'Status' is 'ENTERED'. The 'Usage Lines' table is empty, showing 'There are no rows to display.'

2.a Inventory Usage Tab, filled information about new inventory usage tab

	2.b Usage, running number usage automatically filled by system 2.c Filled field about details Inventory Usage 2.d Filled field with Storeroom listed 2.e Button “Select Items for Return”. Searching item to return.
3	Show all listed item out for warehouse  Checked list item where item return and then click “OK”
4	 4.a Usage lines filled all items choose. 4.b Status Inventory Usage is ENTERED 4.c After all field complete, user can Save this Inventory Usage. In the top page show information “Record has been saved”. 
5	Back to home  menu. Then looked “Material Reservation List”, searching for Inventory usage created before.

Welcome, Storeman01

Storeman01

PO to be Receipts List

PO	Description	Company	Total Cost	Receipts	Ordered Date
1013	UAT Test 01	709831	6,000.00	NONE	2/2/16 6:04 PM
1015	Generated by reorder 2/11/16 4:46 PM	MT SUI	25,000.00	NONE	2/11/16 4:47 PM
1020		MT SUI	1,100.00	NONE	2/22/16 2:34 PM
1021		MT SUI	100.00	NONE	2/22/16 2:41 PM
POINT001	Test Integration 001	709654	70,700,000.00	NONE	2/26/16 1:40 PM
GF SPO1001	Test Integrationload	709654	102,100,000.00	NONE	2/29/16 4:26 PM
GF SPO1008	Test Integrationload	709654	8,000,000.00	NONE	2/29/16 5:02 PM
GF SPO1007	Test Integrationload	709654	100,000,000.00	NONE	2/29/16 5:02 PM
PO12345	blablabla	708163	1,050.00	NONE	3/1/16 9:37 PM
PO44457	blablabla	708163	400.00	NONE	3/3/16 5:46 PM

Set Chart Options 1 - 10 of 41 Next Page >>

Inbox / Assignments

No Assignments found for Storeman01

Material Reservation List

Usage	Description	From Location	Usage Type	Status
1121	Created for WO - 16000478.	NPCT1-ENG-STORE	ISSUE	ENTERED
1122	Created for WO - 16000483.	NPCT1-ENG-STORE	ISSUE	ENTERED
1123	Created for WO - 16000488.	NPCT1-ENG-STORE	ISSUE	ENTERED
1159	Inventory Usage Return Sample 01	NPCT1-ENG-STORE	MIXED	ENTERED
1160	Inventory Usage Return Sample 02	NPCT1-ENG-STORE	MIXED	ENTERED
1161	Inventory Usage Sample Return 03	NPCT1-ENG-STORE	MIXED	ENTERED
1155	Created for WO - 16000502.	NPCT1-ENG-STORE	ISSUE	ENTERED
1146	Open Record Created for WO - 1032.	NPCT1-ENG-STORE	ISSUE	ENTERED
1131	Created for WO - 16000528.	NPCT1-ENG-STORE	ISSUE	ENTERED
1132	Created for WO - 16000532.	NPCT1-ENG-STORE	ISSUE	ENTERED

Set Chart Options Previous Page 101 - 110 of 117 Next Page >>

PR need to be Submit to Approval

PR	Description	Required Date	Total Cost	Status
1017	Lamp Warning		600,000.00	WAPPR
1018	Pump 2	5/5/16 10:37 AM	10,000.00	WAPPR
PRENG1002416/EMS	Test		0.00	WAPPR

6 Open Inventory Usage,

Query Find Usage Select Action

List View Inventory Usage

Usage: 1161 Inventory Usage Sample Return 03 Usage Type: MIXED Attachments

From Storeroom: NPCT1-ENG-STORE Engineering Storeroom Site: NPCT1 Status: ENTERED

Inventory Owner:

Usage Lines 1 - 1 of 1

Line	Item	Description	From Bin	Usage Type	Quantity	Split?
1	ITEM002	Item test integration10		RETURN	1.00	

Select Items Select Items for Return Select Reserved Items Select Asset Spare Parts New Row

6.a Information status ENTERED

6.b Change status to Complete, show dialog box

Change Status

Usage: 1161 Inventory Usage Sample Return 03

Status: ENTERED Entered

New Status: Complete

Memo: Complete Return Inventory Usage 1161

OK Cancel

Filled New Status Complete and filled memo to reason or information complete. Finally, click

“OK’ to progress, after that Status Inventory Usage change to COMPLETE automatically.

[List View](#) **Inventory Usage**

Usage: 1161 [Inventory Usage Sample Return 03](#)  Usage Type: MIXED  [Attachments](#)

From Storeroom: NPCT1-ENG-5 >> Engineering Storeroom  Site: NPCT1 **Status: COMPLETE**

Inventory Owner: 

2.6 Exercise Phase